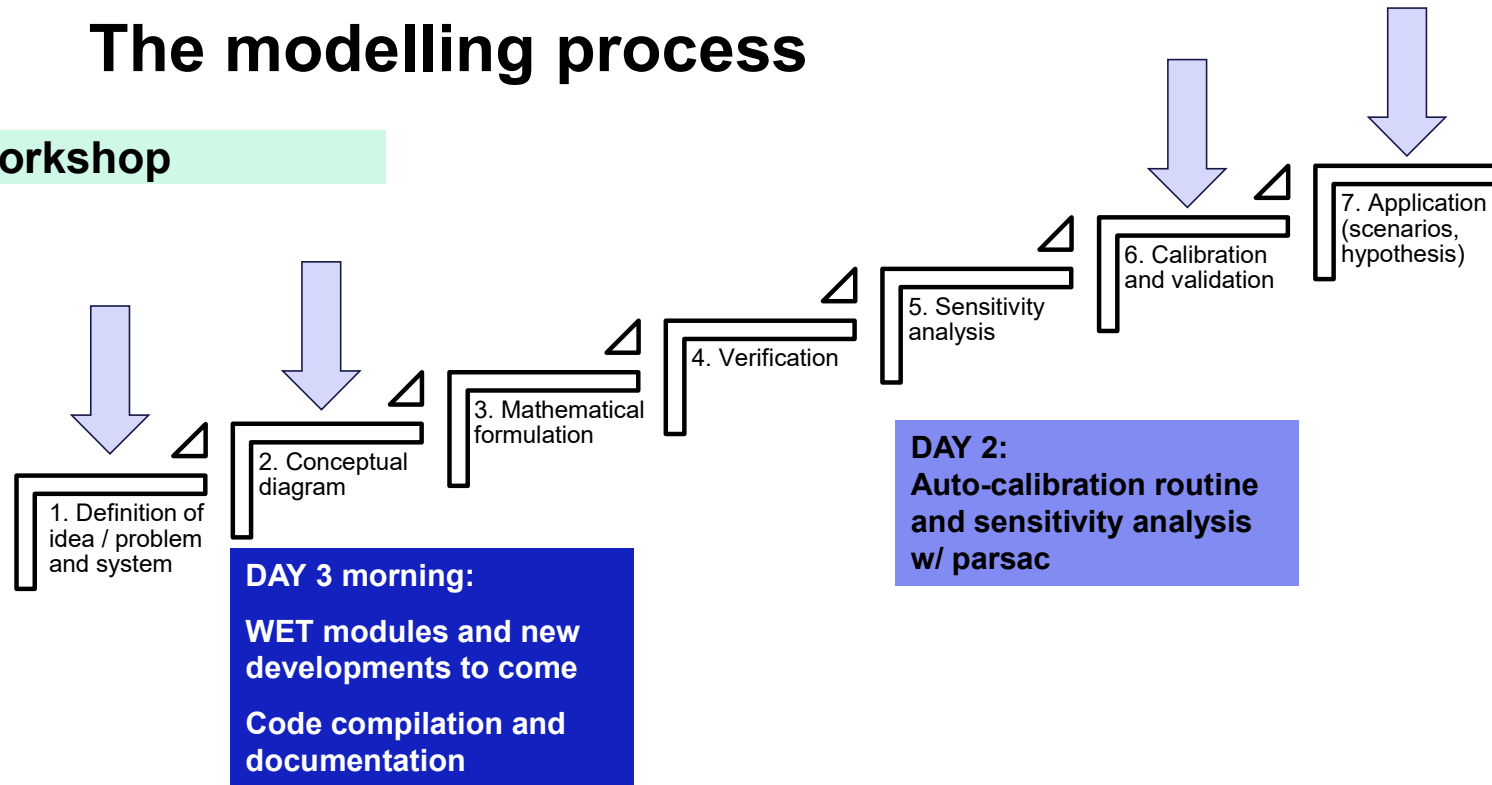


Day 2	
9:00-9:15	Catch-up on the previous day
9:15-10:15	Lecture 3A: Calibrating lake ecosystem models: Recommendations for WET users
10:15-10:30	Break
10:30-11:00	Tutorial 2: Introduction to parsac and auto-calibration
11:00-12:00	Hands-on exercise 2: Auto-calibration of your WET model
12:00-13:30	Break
13:30-14:15	Lecture 3B: Evaluating lake ecosystem models: Recommendation for WET users
14.15-14.45	Break
14.45-16.00	Continue and finish exercise 2
16.00-17.30	Q&A

The modelling process

This workshop



DAY 1: QWET exercise – from scratch to model set-up, simulation and scenarios

Workflow in QWET



1. Model configuration and setup
2. Parameter adjustments
3. Model simulation
4. Model output visualization
- including comparison with observations
5. Scenario design
6. Scenario simulation
7. Scenario output visualization and evaluation

RECAP: WET exercise

Food web modules

Fish

Zooplankton

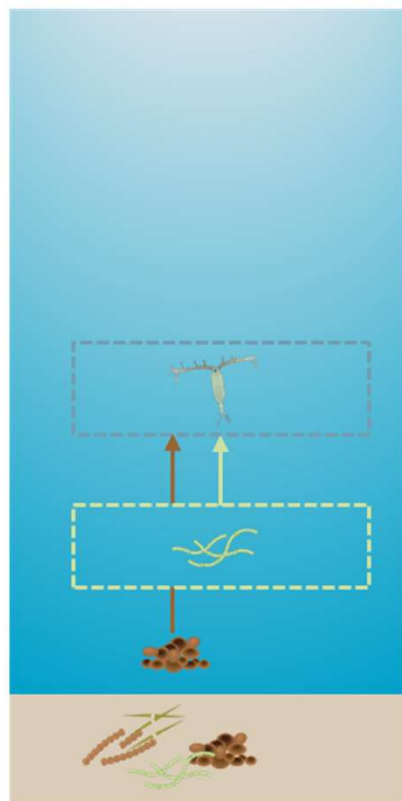
Phytoplankton

Macrophytes

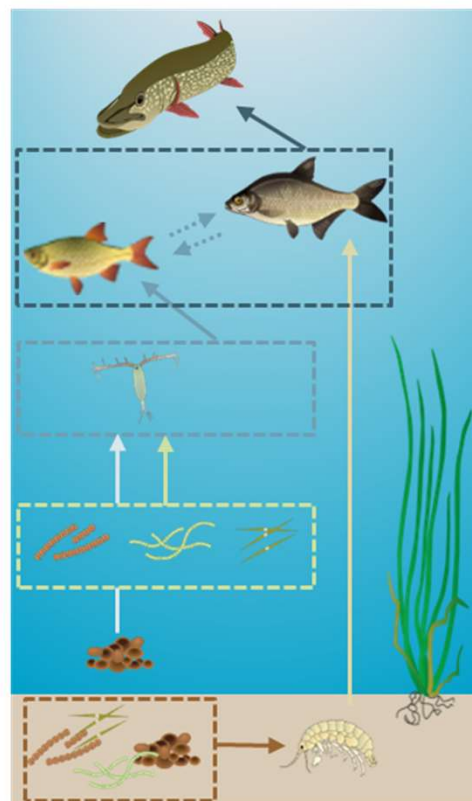
Detritus

Zoobenthos

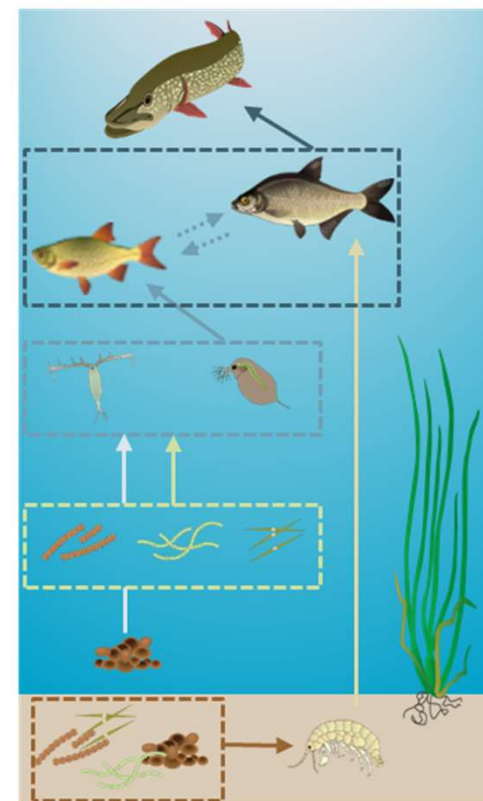
NPZD



WET standard

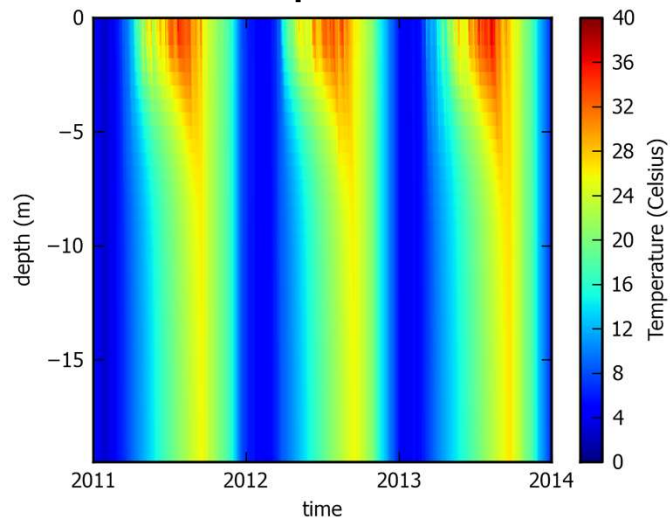


WET advanced

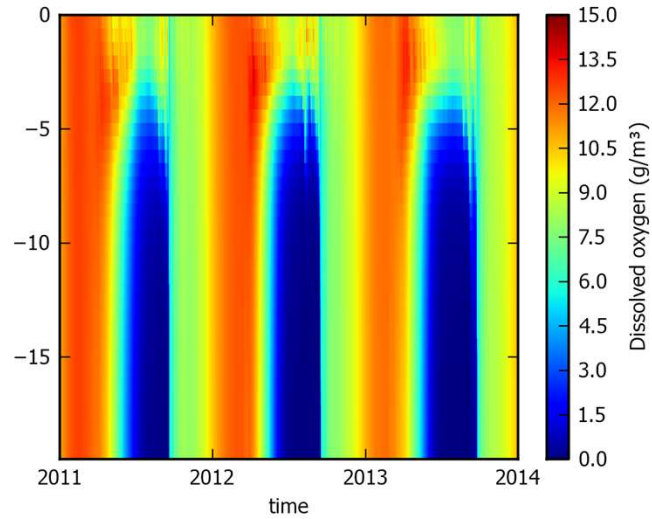


RECAP: WET exercise 1

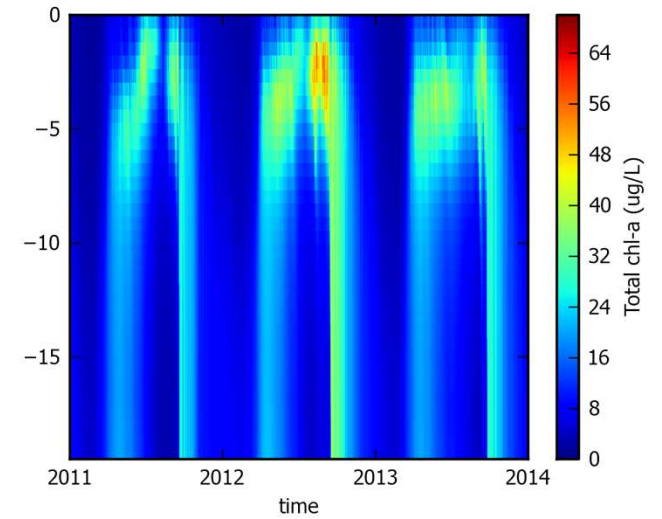
Temperature



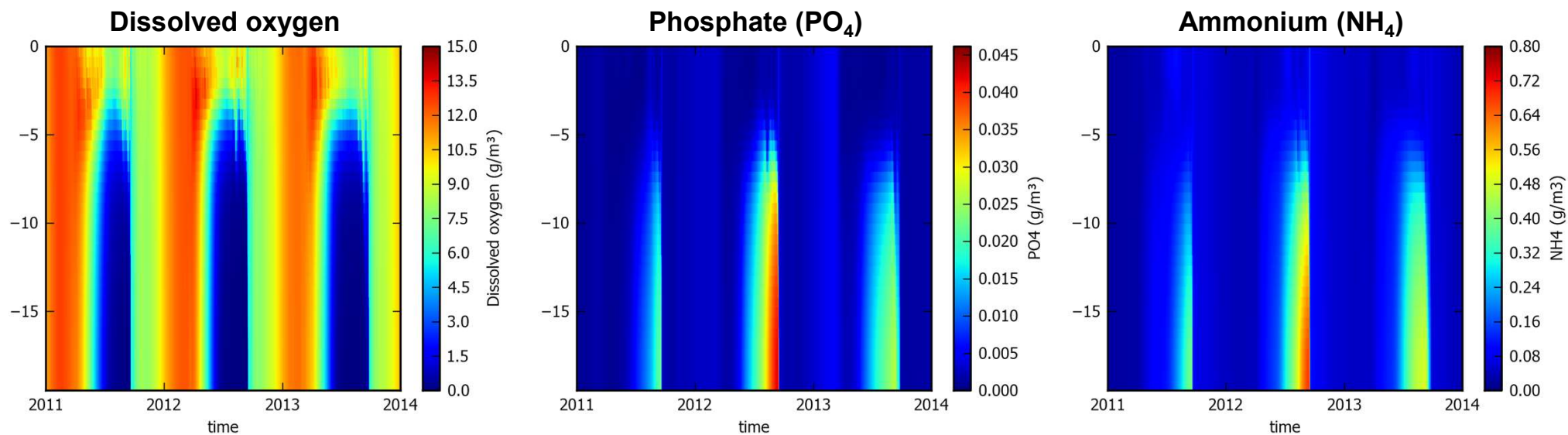
Dissolved oxygen



Total chl-a

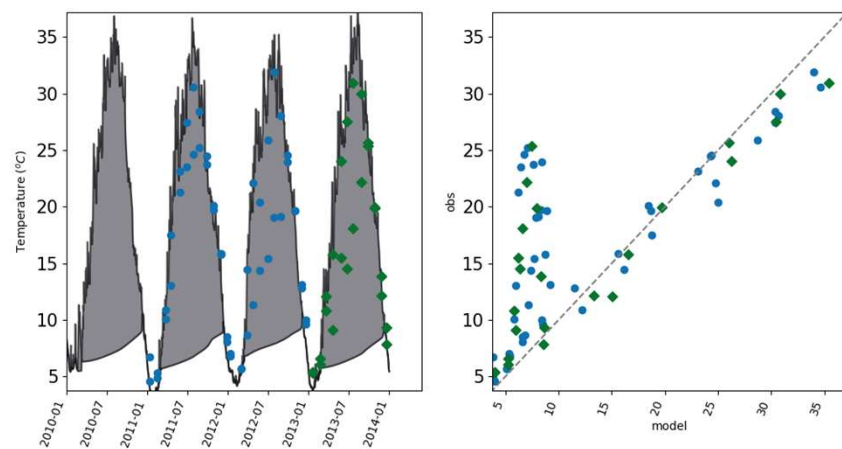


RECAP: WET exercise 1

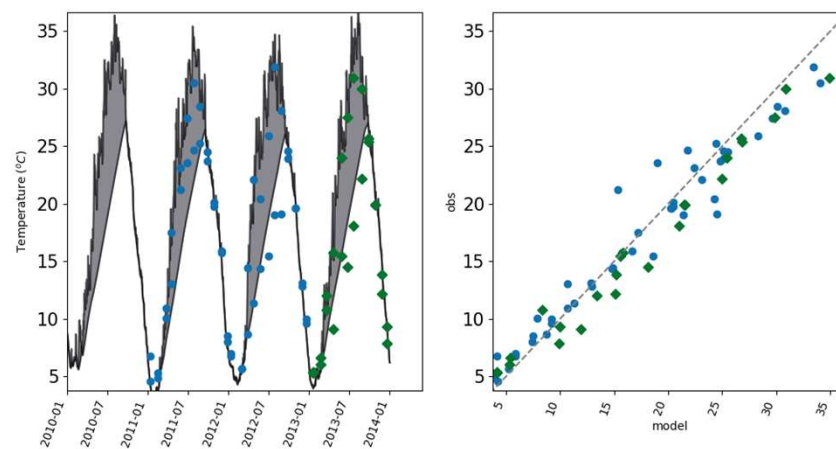


DTU Water temperature calibration

Default set-up

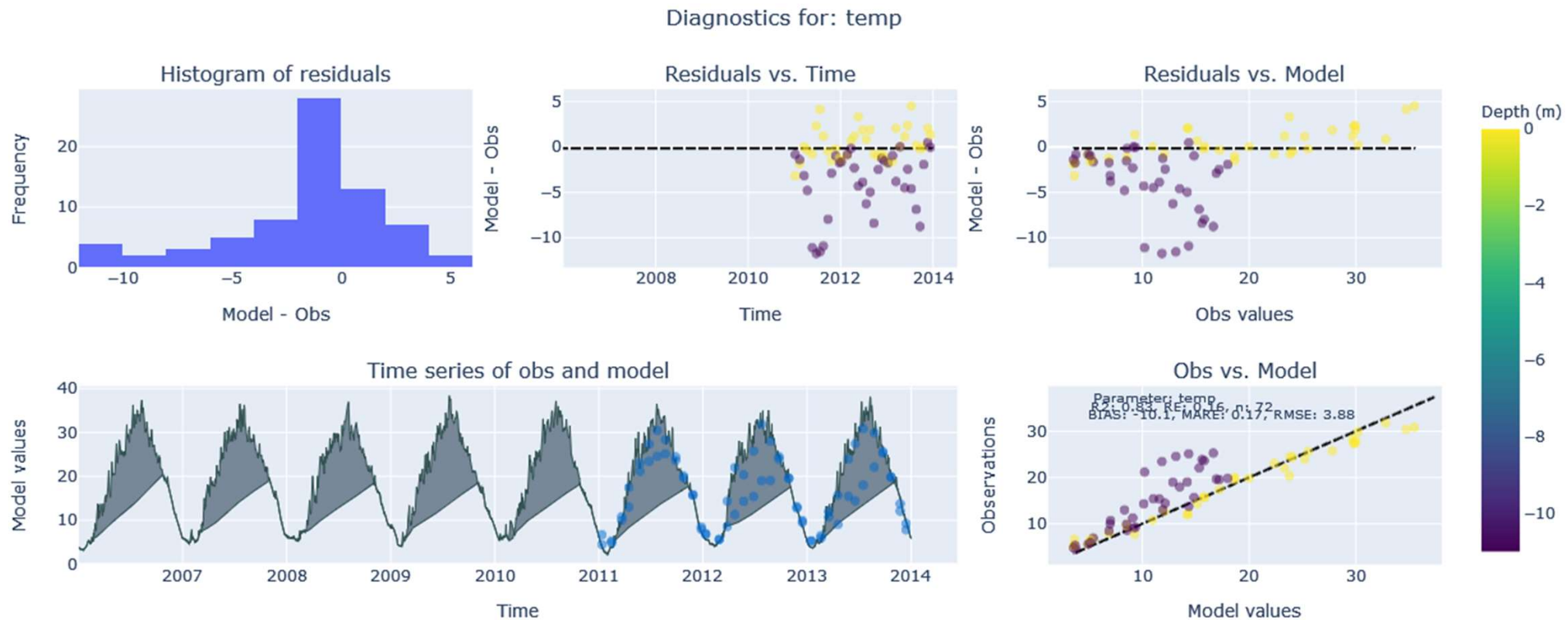


After $k_{\min}$ calibration



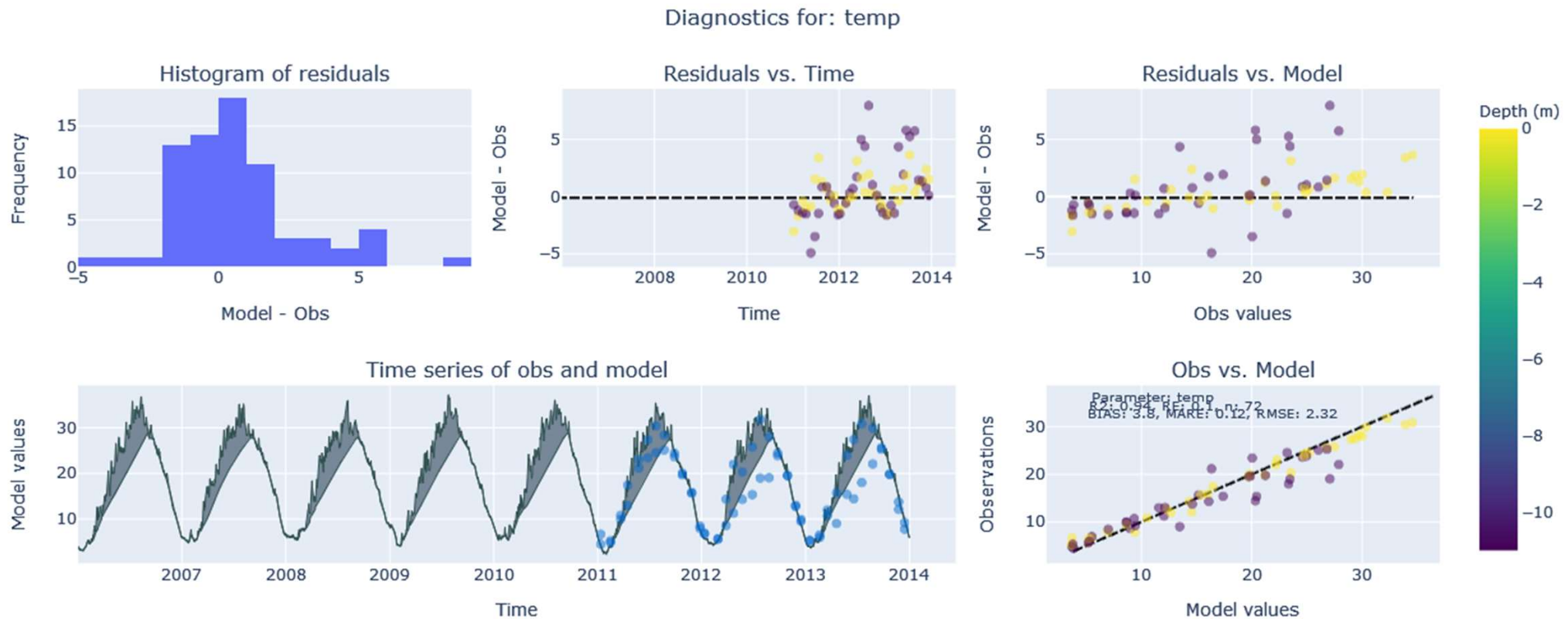
DTU In-depth model performance diagnostics

Temperature (uncalibrated)

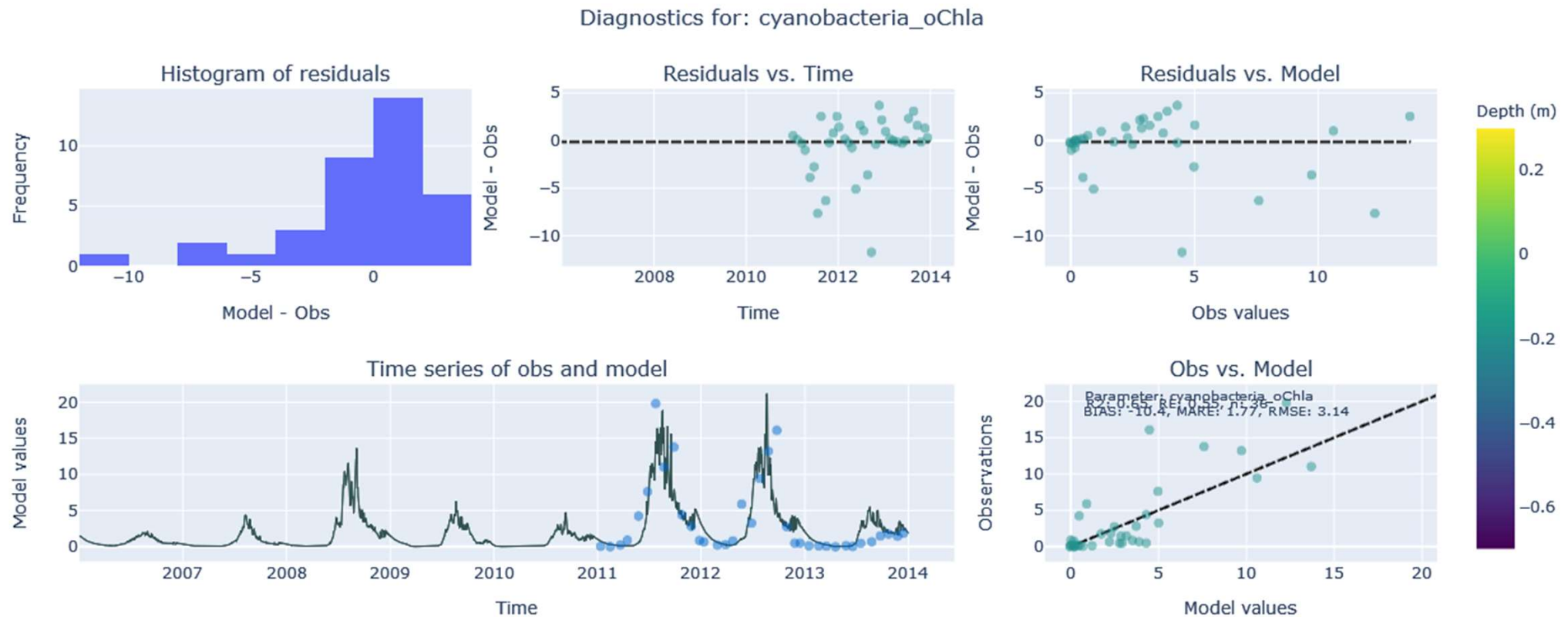


DTU In-depth model performance diagnostics

Temperature (calibrated)

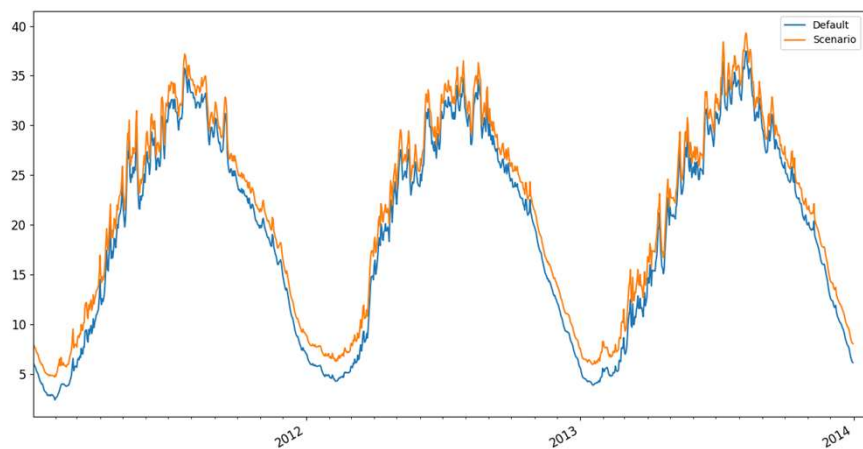


DTU Cyanobacteria chl-a (calibrated)

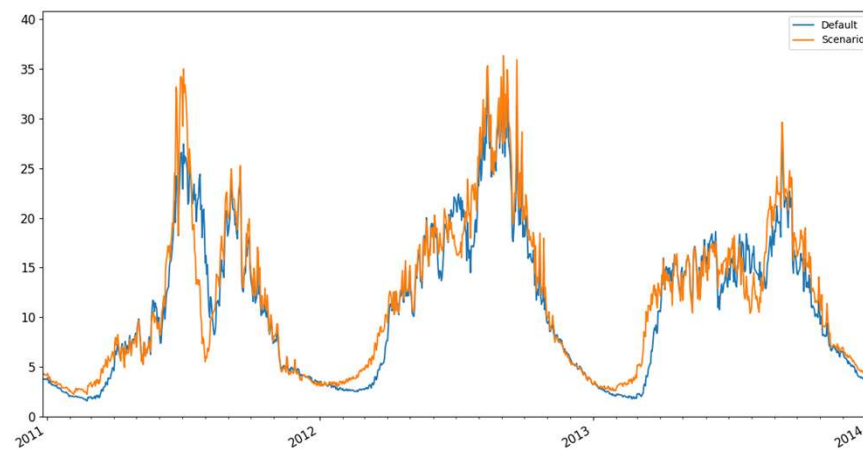


Climate warming scenario

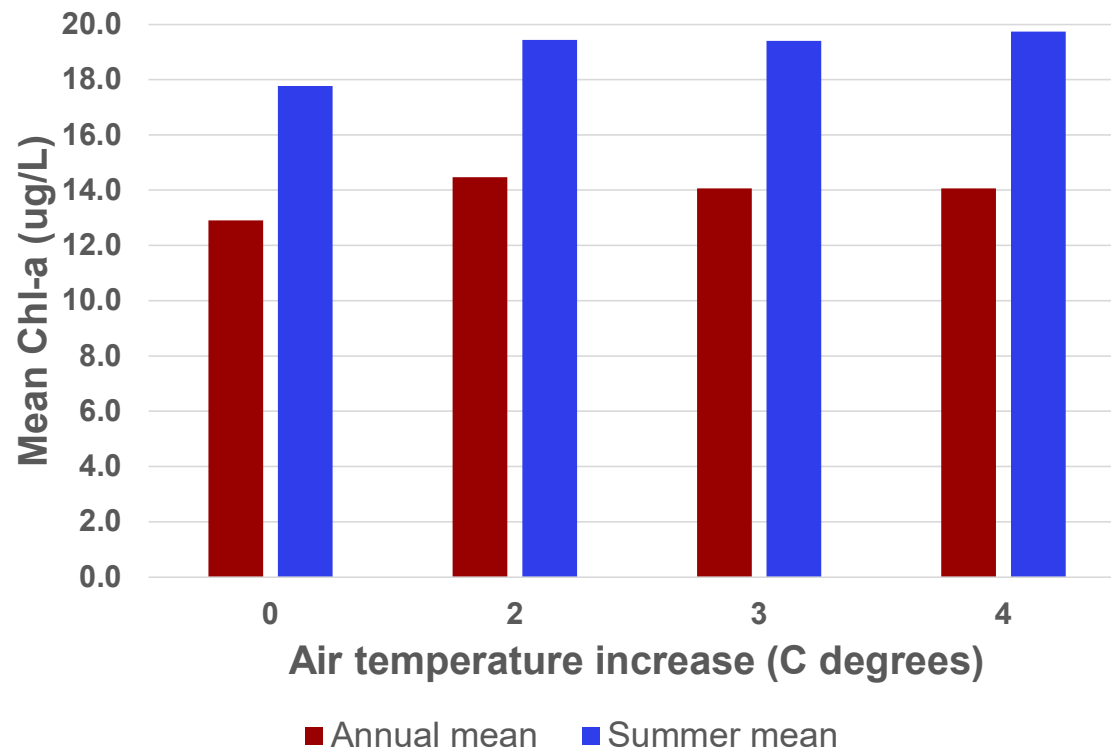
Temperature



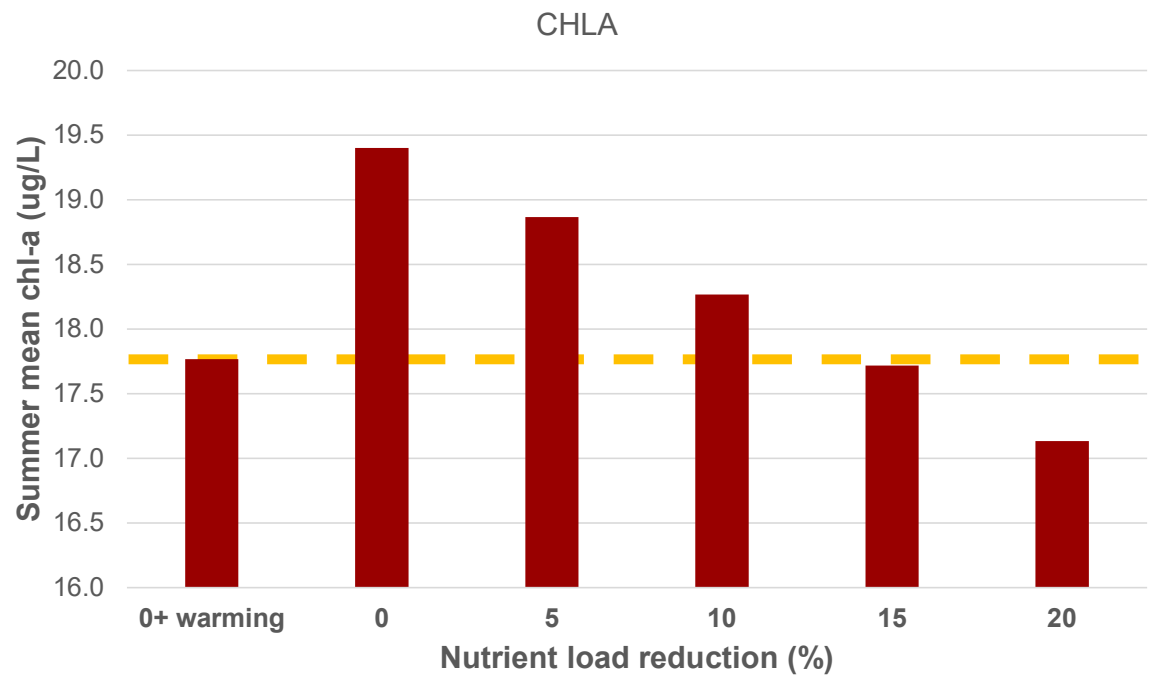
Total chl-a (ug/L)



DTU Total chl-a response to warming



How much should we reduce external nutrient load to counteract climate warming of 3 °C?





Tobias K Andersen, Xiangzhen Kong

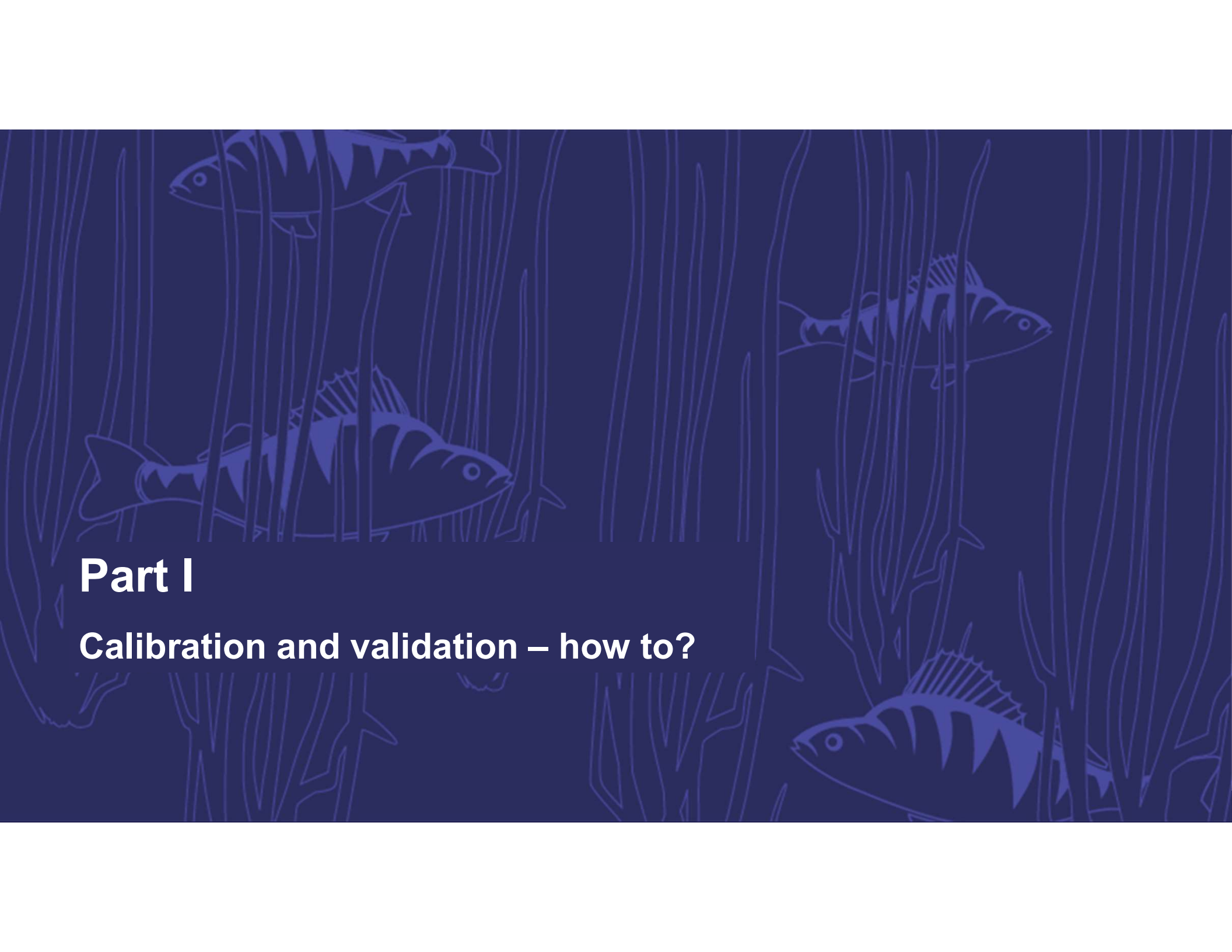
Calibrating lake ecosystems: Recommendations for WET users

Aim of presentation

Learning outcomes:

Participants will get an overview of

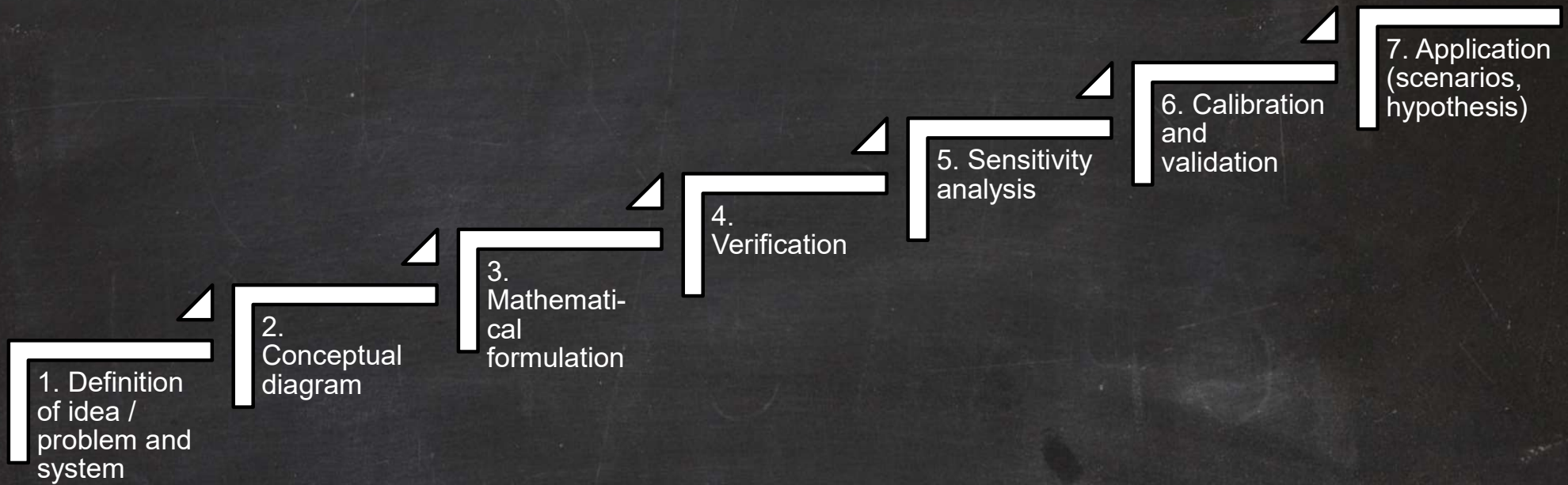
- Calibration and validation process
- Metrics and plots for model performance
- Strategies for the calibration process
- Specific WET model considerations
- How to use SA results for the calibration
- Framework for lake model evaluation

A stylized illustration of four striped fish, possibly perch, swimming through tall, thin reeds in a pond. The fish are depicted in a simple, graphic style with dark stripes on a lighter body. The reeds are represented by thin, vertical lines. The background is a solid dark blue color.

Part I

Calibration and validation – how to?

The modelling process





We don't all agree: Varying terminology for model evaluation

- Model verification
 - Model validation
 - Model evaluation
-
- Terminology changes with discipline, field and modelling context.
-
- In process-based aquatic ecosystem model often:
 - Code verification
 - Model validation (in combination w/ model calibration)
 - Model evaluation:
Overall process of investigating model simulation output



Why is model validation important?

- Models must be validated for several reasons
 - Increase scientific credibility
 - Detect errors
 - Assess accuracy
 - Determine the validity range of the model
- All scientific models are capable of being validated (falsified)
- But the specific validation practices depends on model type and purpose of the model

Calibration and validation

6. Calibration and validation

“**Calibration** is adjusting model parameters with the purpose of achieving the best simulation match with observations.”

This may be achieved by a combination of visual inspection and optimization of an objective function.

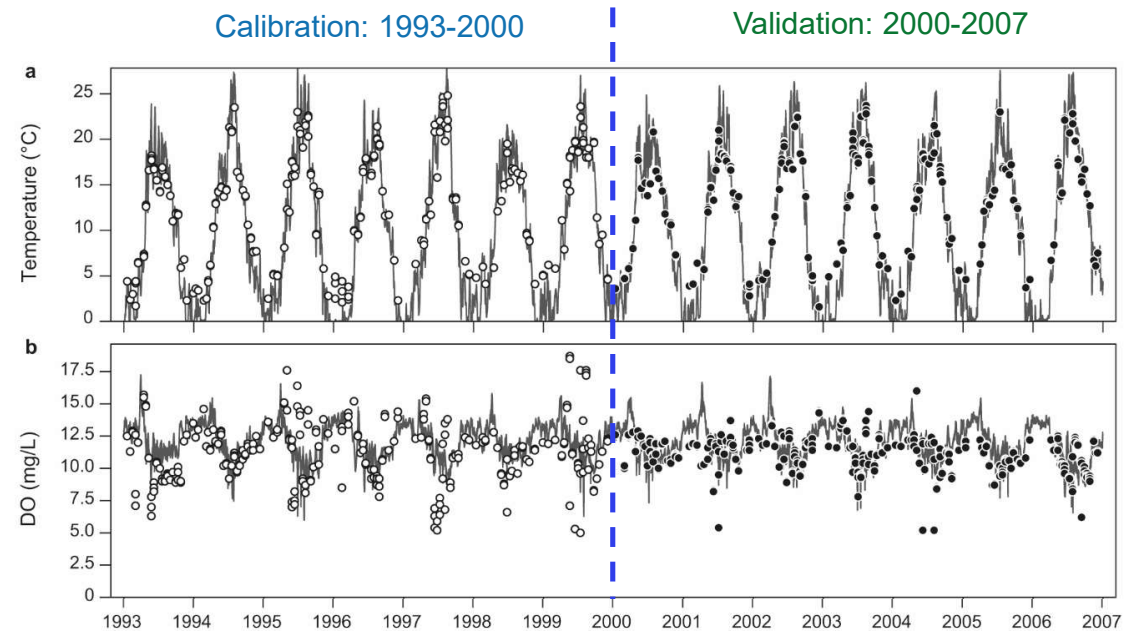
For example, minimizing the Mean Absolute Percentage Error (MAPE):

$$MAPE(Chla) = \frac{1}{n} \sum_{i=1}^n \frac{|Chla_{simulated} - Chla_{observed}|}{Chla_{observed}}$$

6. Calibration and validation

“**Validation** is the process of testing the calibrated parameters with an independent set of data (in time and/or space)”

Model adjustment to Lake Hinge case

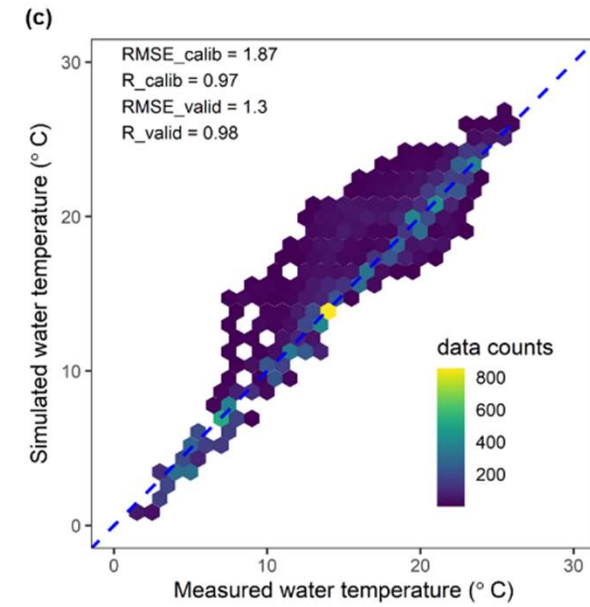
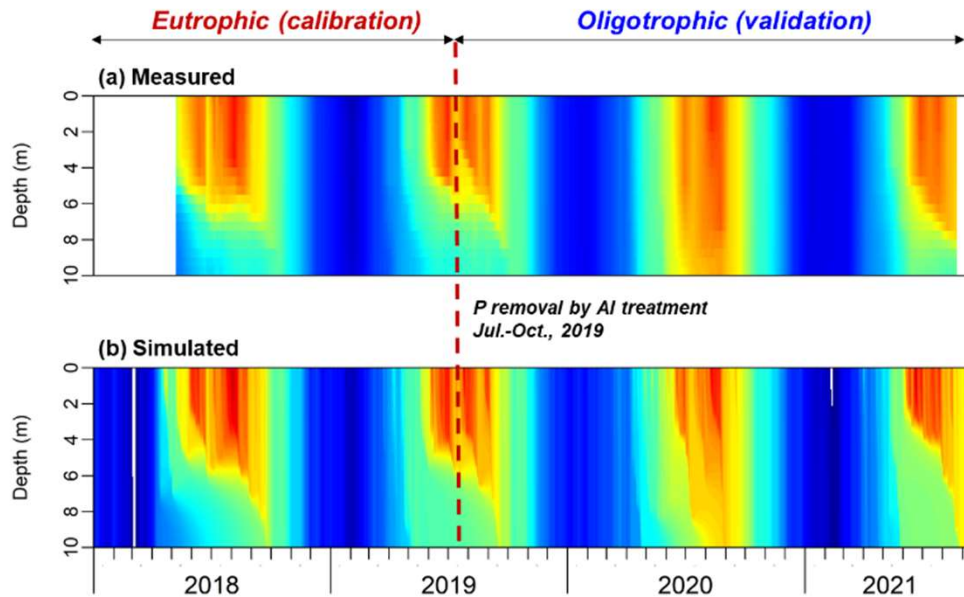


Andersen et al, 2020, *Ecol. Appl.*

Calibration and validation

6. Calibration and validation

Model adjustment to Lake Barleber case



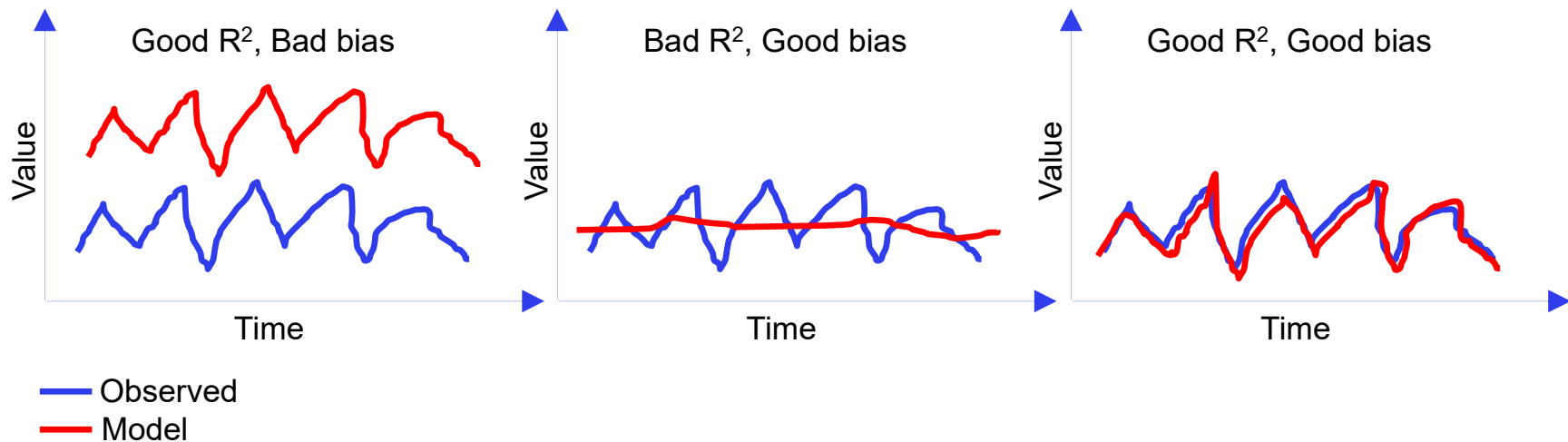
Kong et al., ES&T 2023

6. Calibration and validation

Choosing objective functions (model performance metrics) for calibration and validation

The different types of model performance metrics have both strengths and weaknesses. Multiple performance metrics should therefore always be evaluated – ideally combining residual and correlation type metrics. For example, the coefficient of determination r^2 , which can suffer significant offset error, could be paired with percent bias (Bennett et al. 2013).

“an **objective function** is a mathematical expression for how closely model output matches observations”

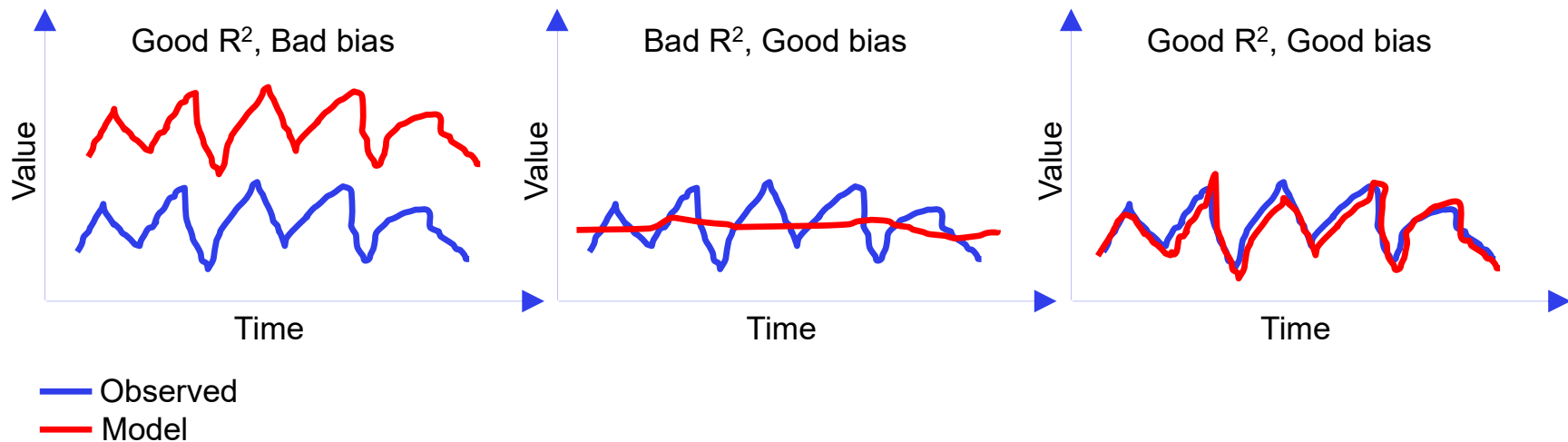


6. Calibration and validation

Choosing objective functions (model performance metrics) for calibration and validation

The different types of model performance metrics have both strengths and weaknesses. Multiple performance metrics should therefore always be evaluated – ideally combining residual and correlation type metrics. For example, the coefficient of determination r^2 , which can suffer significant offset error, could be paired with percent bias (Bennett et al. 2013).

“an **objective function** is a mathematical expression for how closely model output matches observations”

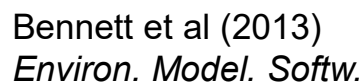


Choosing objective functions (model performance metrics) for calibration and validation

Examples of commonly used **residual performance metrics**:

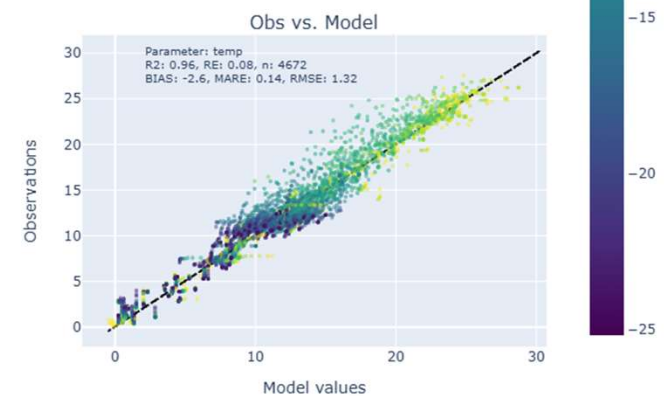
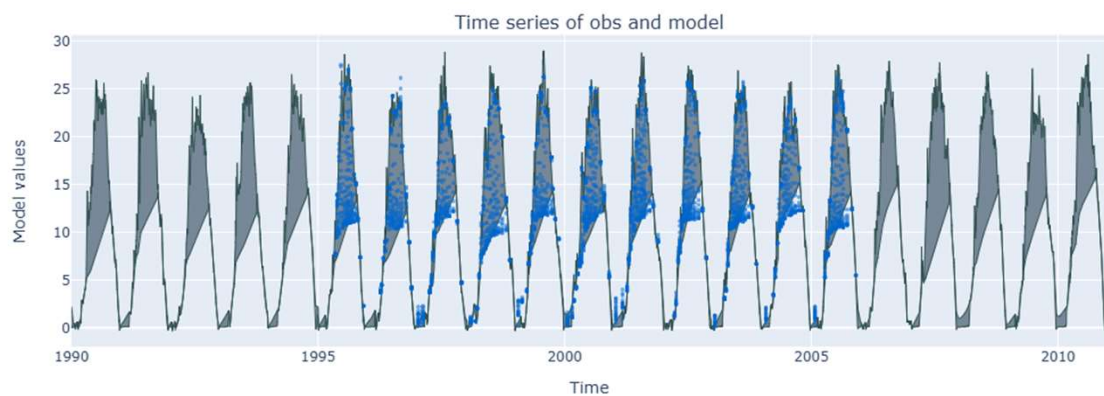
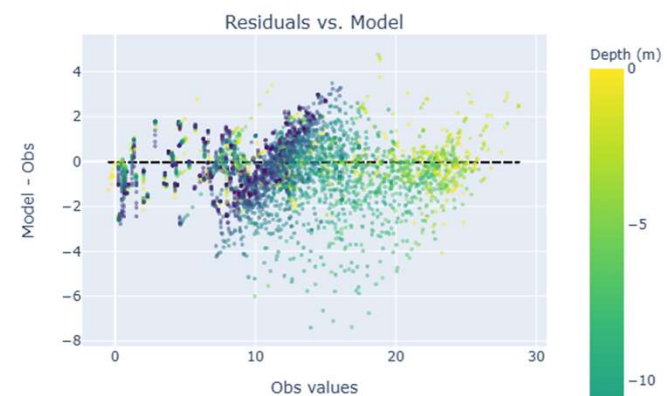
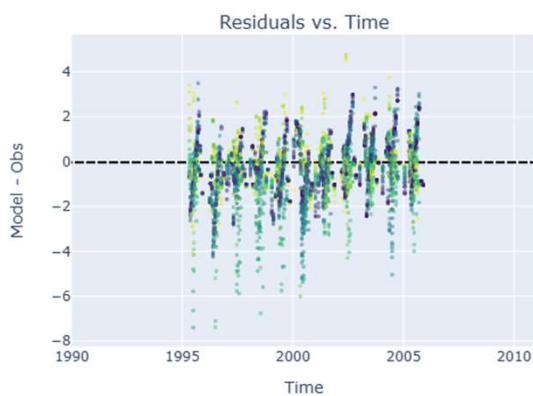
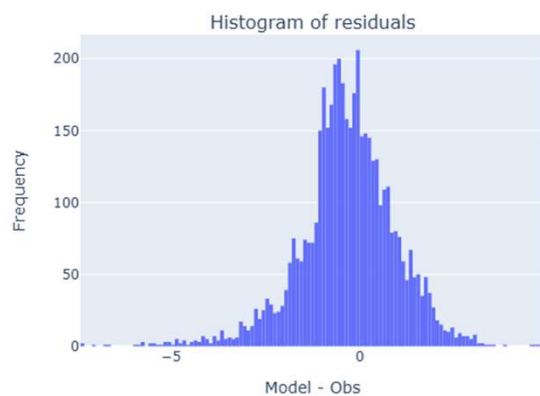
- Examples of commonly used **correlation and model efficiency performance metrics**:

- “an **objective function** is a mathematical expression for how closely model output matches observations”



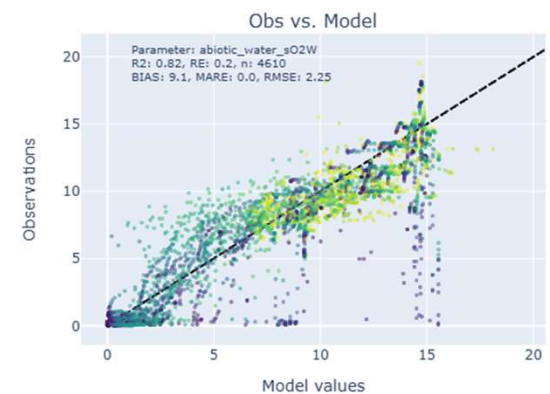
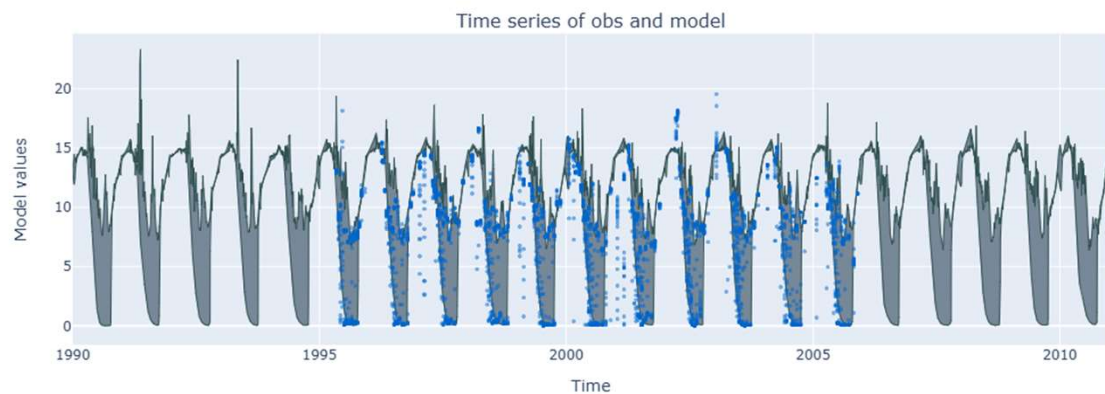
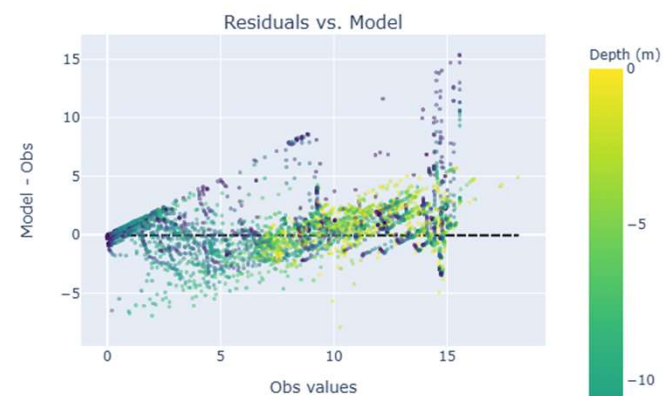
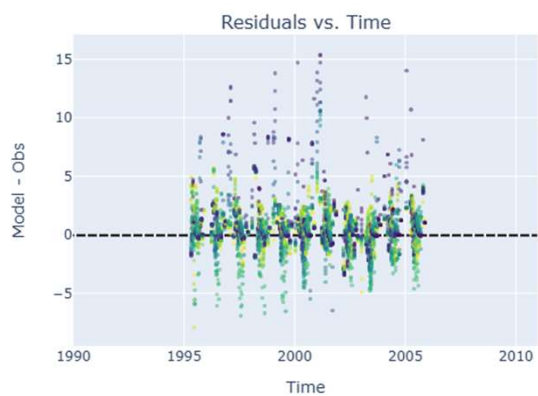
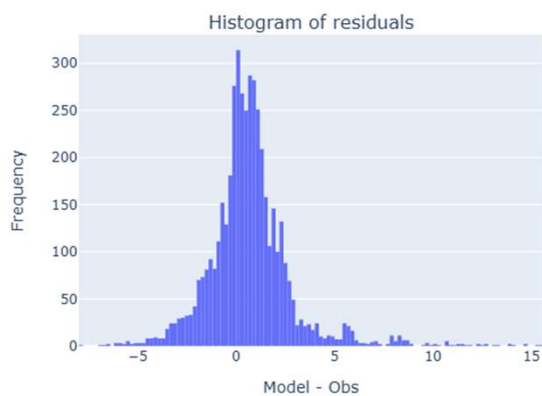
Performance visualizations

Diagnostics for: temp



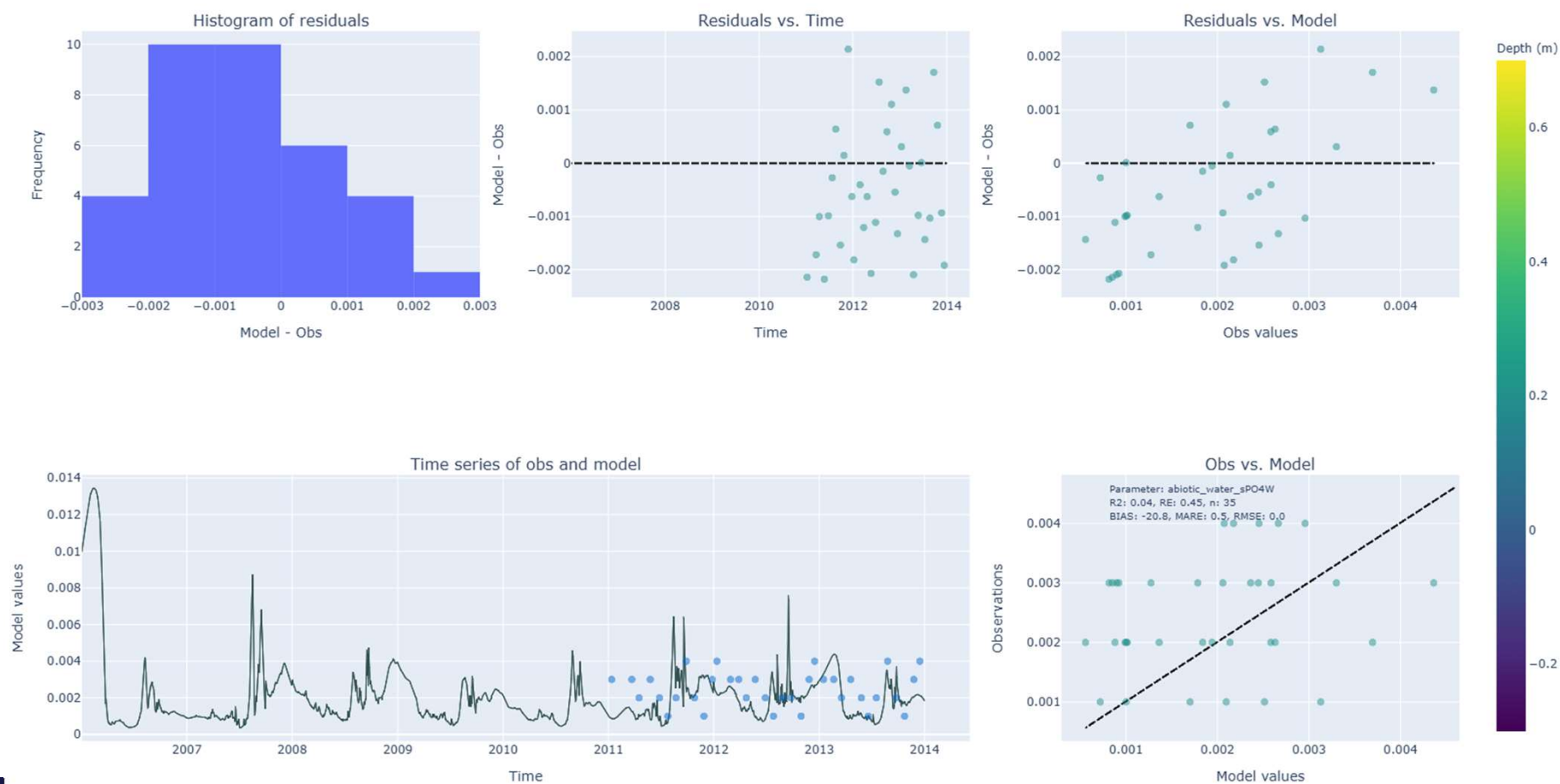
Performance visualizations

Diagnostics for: abiotic_water_sO2W

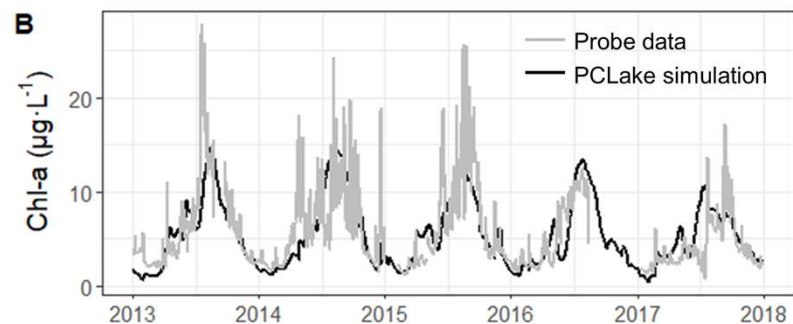
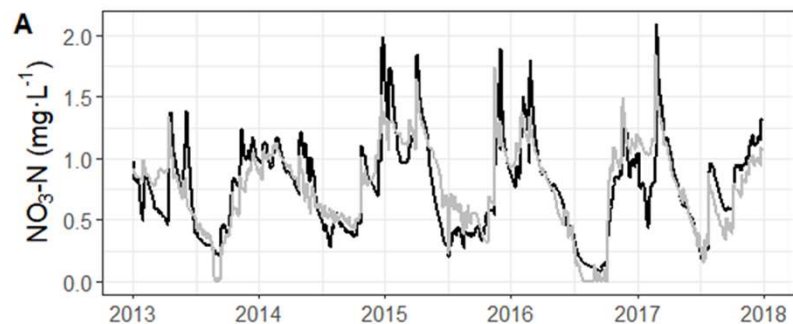


Performance visualizations

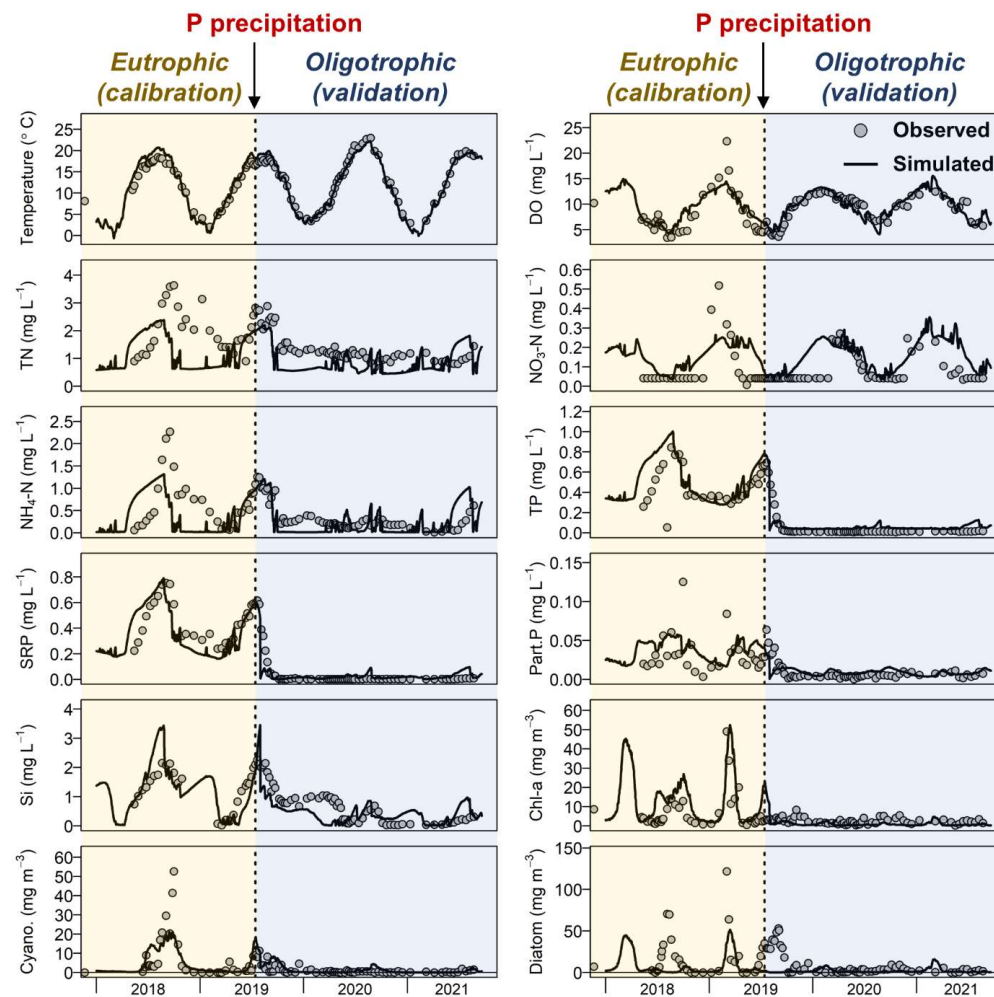
Diagnostics for: abiotic_water_sPO4W



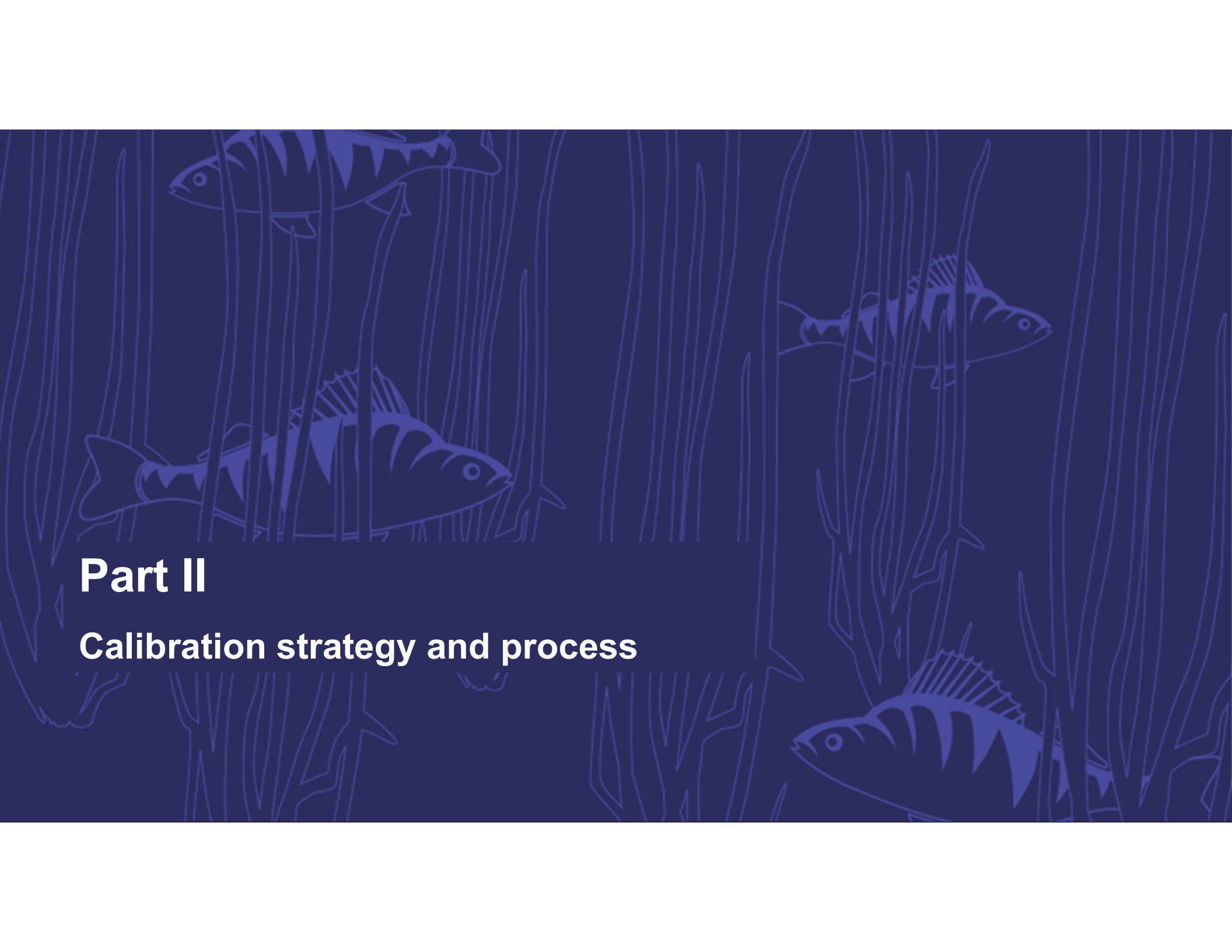
Performance visualizations



Kong et al., Water Res. 2019



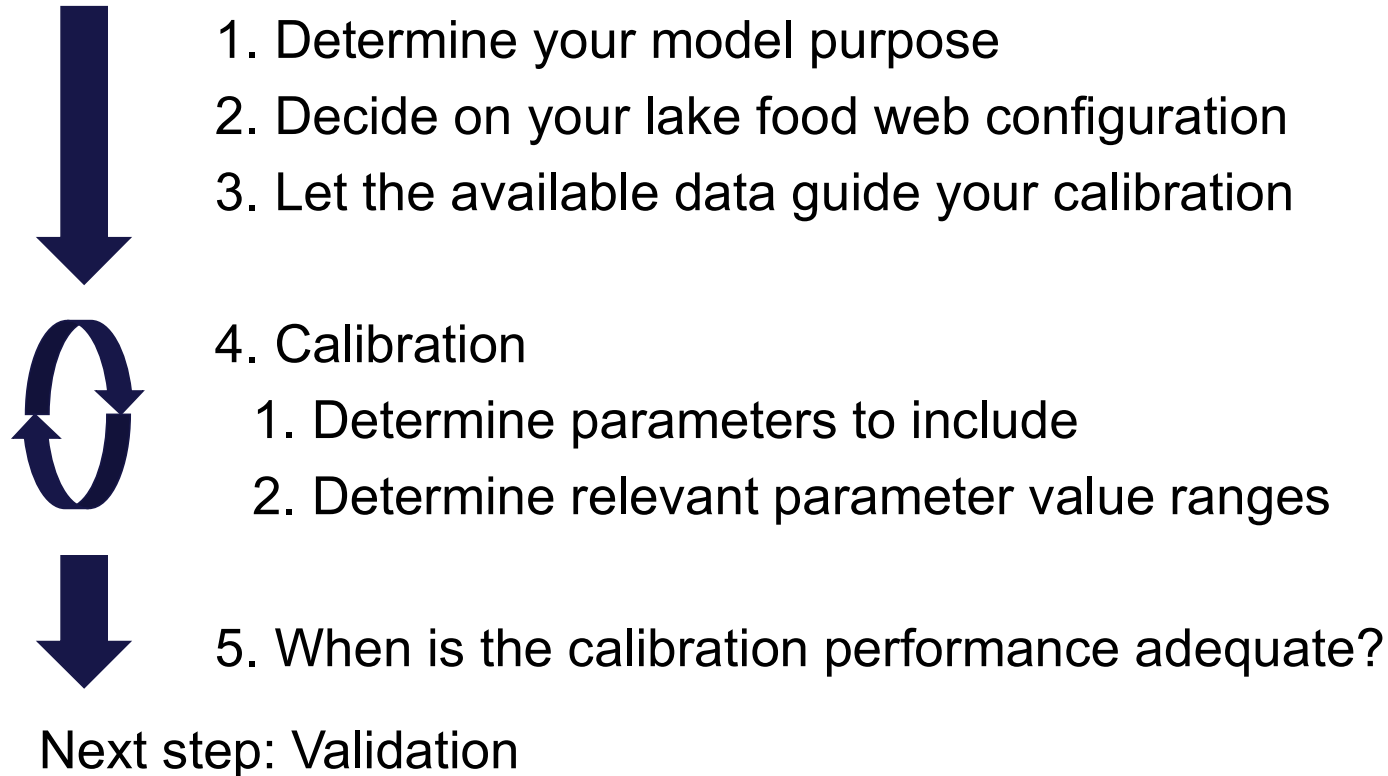
Kong et al., ES&T 2023

A stylized illustration of four striped fish, possibly perch, swimming through tall, thin reeds in a pond. The fish are depicted in a simple, graphic style with dark stripes on a lighter body. The reeds are represented by thin, vertical lines. The background is a solid dark blue color.

Part II

Calibration strategy and process

CSPS framework and model application

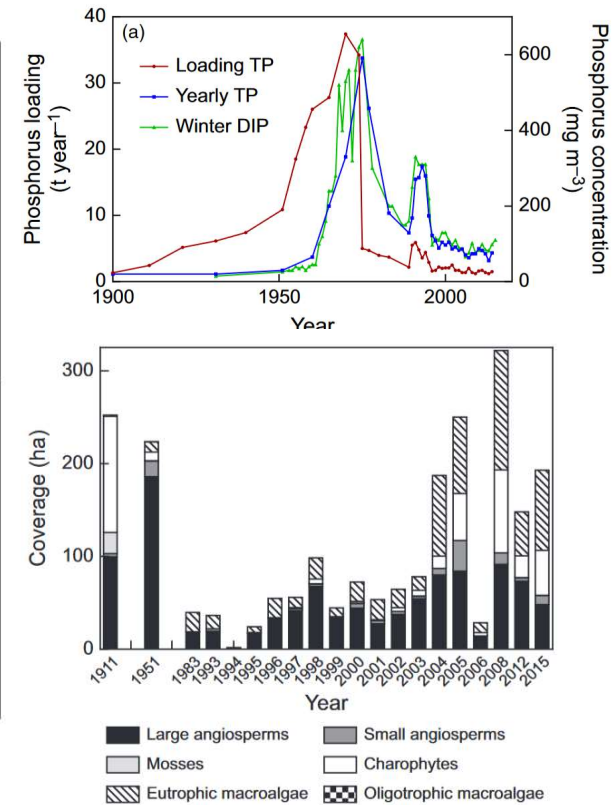
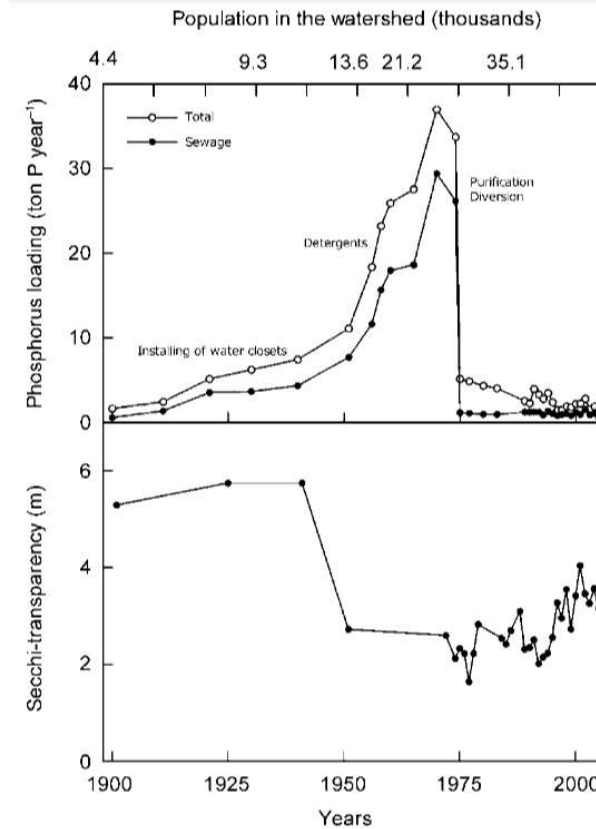


Calibration strategy for WET

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO ₃ and NH ₄
4	Phosphorous	P sediment release	PO ₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		

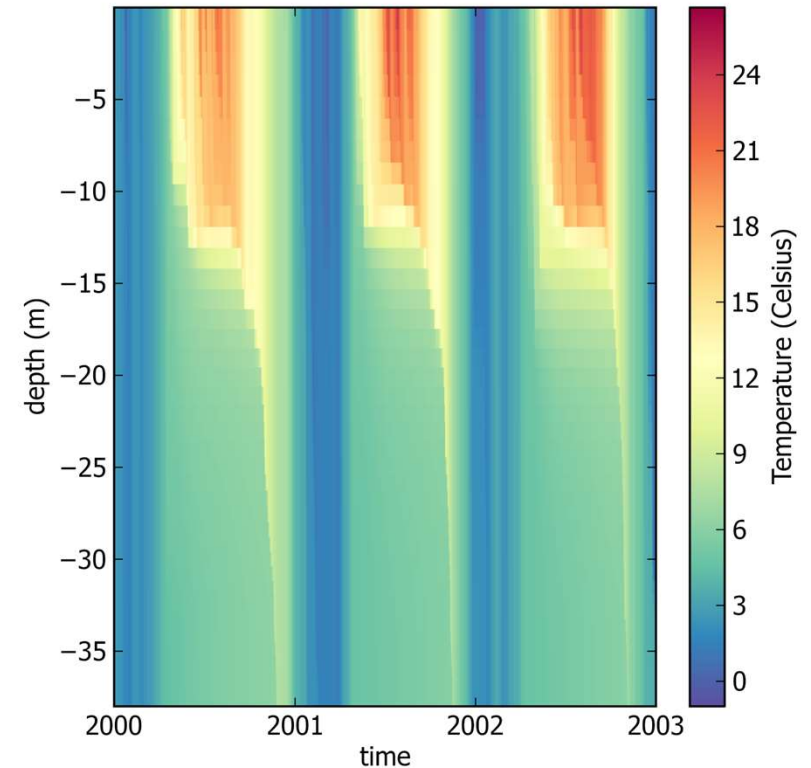


Calibration strategy for WET: Lake Fure, an example



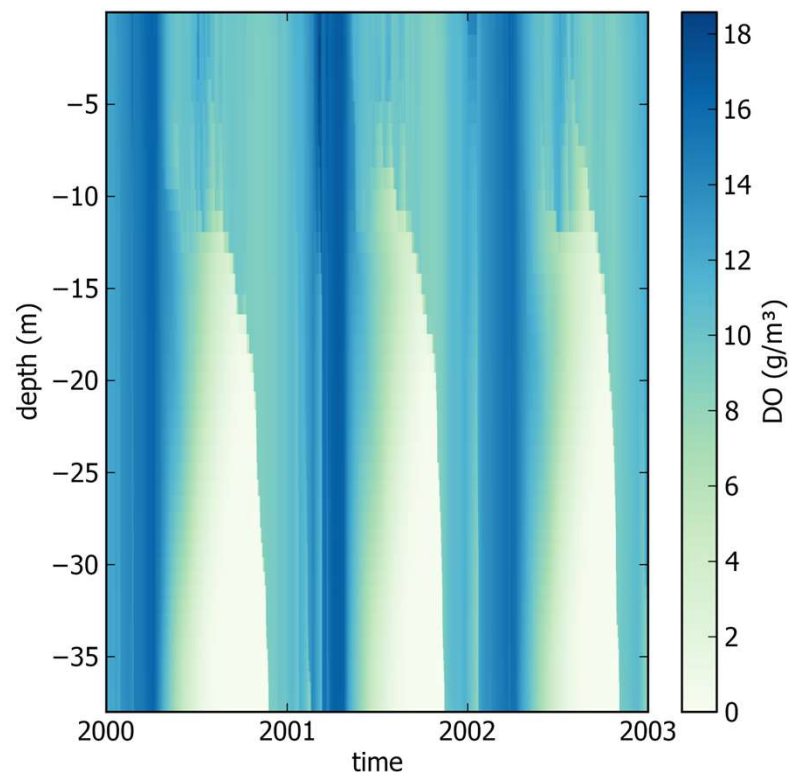
Calibration strategy for WET: Lake Fure, an example

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO ₃ and NH ₄
4	Phosphor	P sediment release	PO ₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		



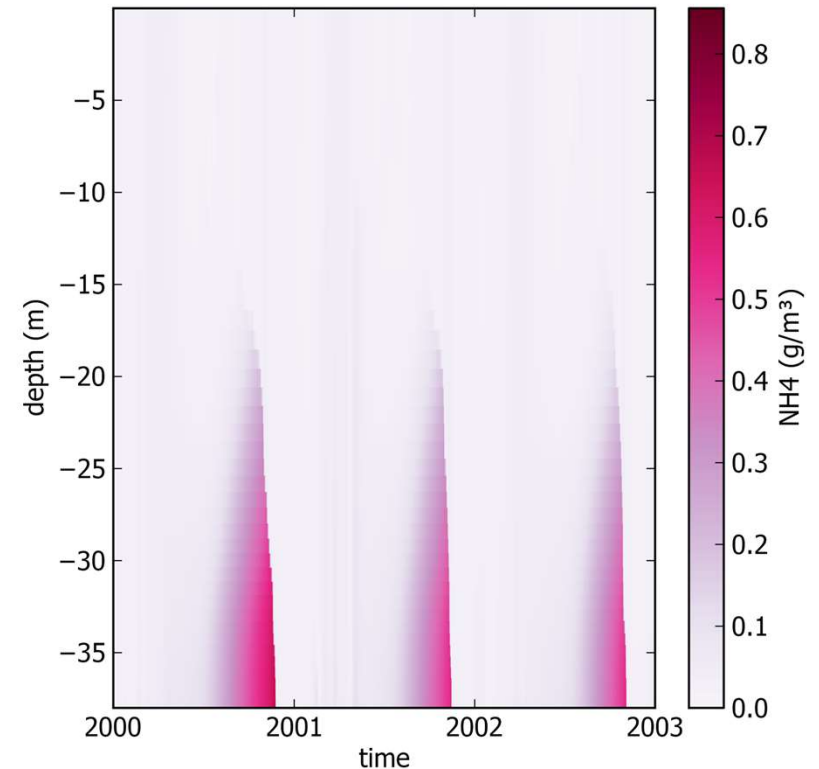
Calibration strategy for WET: Lake Fure, an example

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO_3 and NH_4
4	Phosphor	P sediment release	PO_4
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		



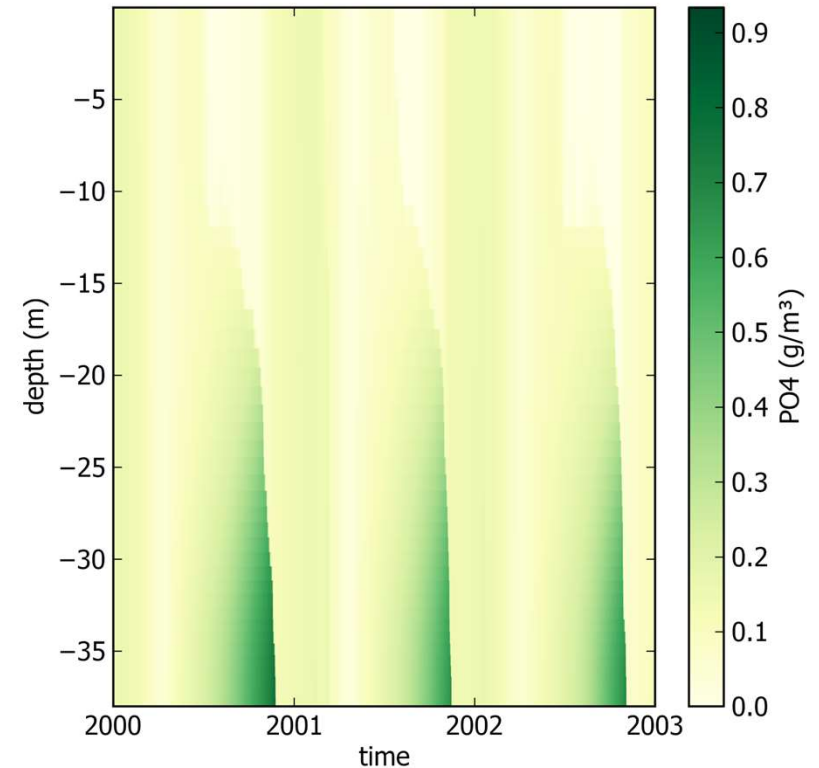
Calibration strategy for WET: Lake Fure, an example

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO₃ and NH₄
4	Phosphor	P sediment release	PO ₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		



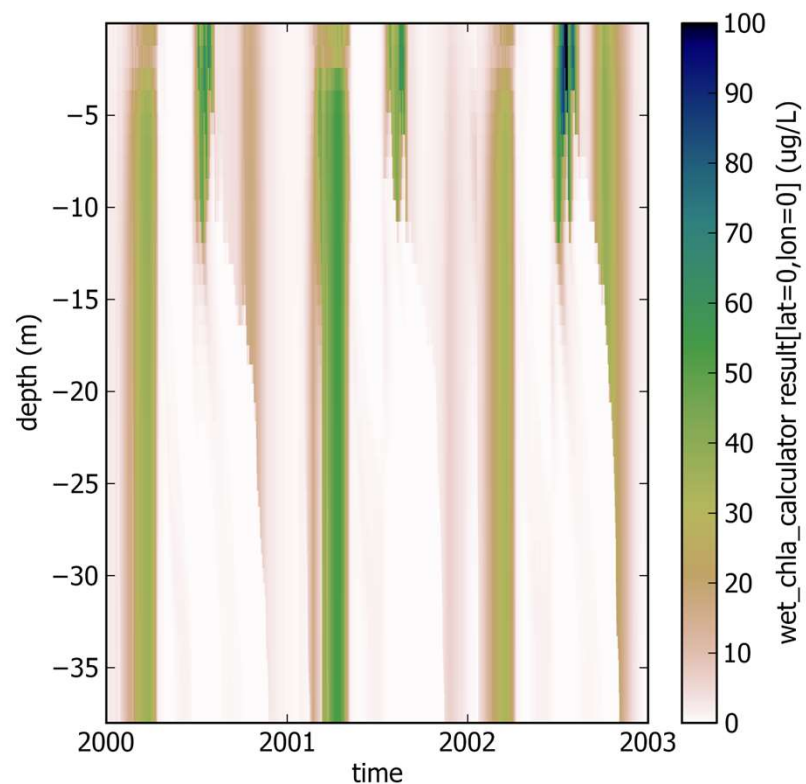
Calibration strategy for WET: Lake Fure, an example

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO ₃ and NH ₄
4	Phosphor	P sediment release	PO₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		



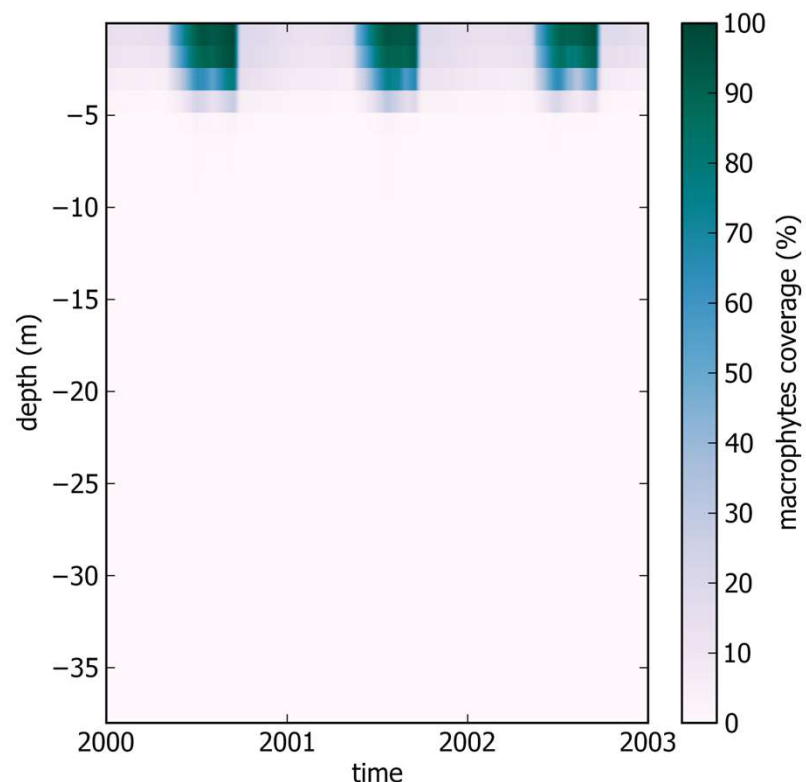
Calibration strategy for WET: Lake Fure, an example

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO ₃ and NH ₄
4	Phosphor	P sediment release	PO ₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		



Calibration strategy for WET: Lake Fure, an example

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO ₃ and NH ₄
4	Phosphor	P sediment release	PO ₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Vegetation	Macrophyte growth	Vegetation coverage





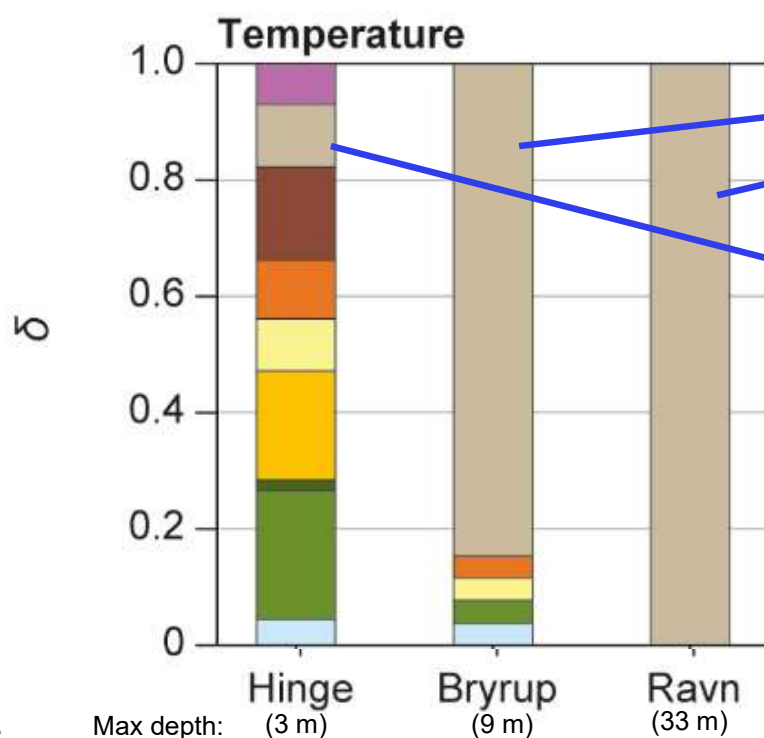
Selecting important model parameters

To gather information on which parameters to include in your calibration, you should search for

- Sensitivity studies with GOTM, FABM-PCLake and WET
- Model papers with GOTM, FABM-PCLake and WET
- User experiences

Selecting important model parameters

Abiotic_water	Phytoplankton_sediment	Zoobenthos	Wind_factor
Abiotic_sediment	Macrophytes	Fish	K_min
Phytoplankton_water	Zooplankton	Auxiliary	Light_factors

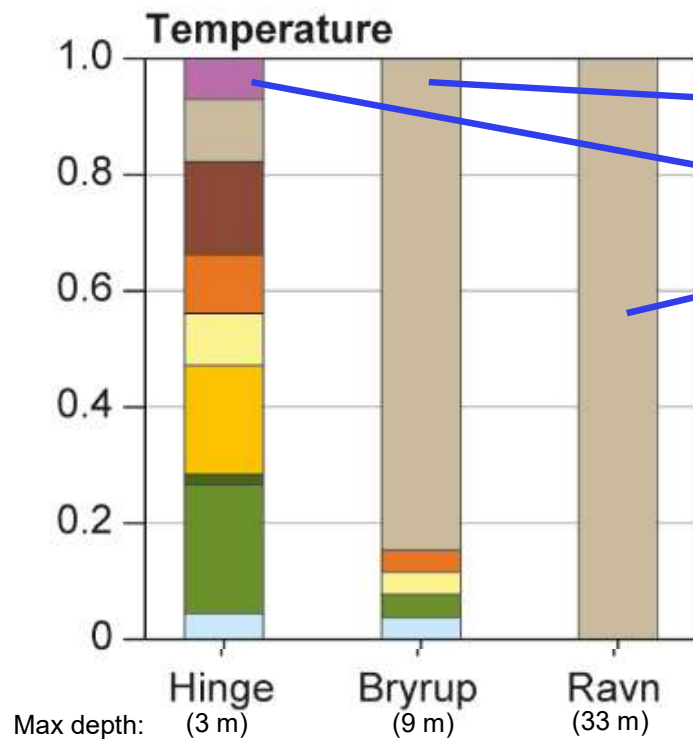
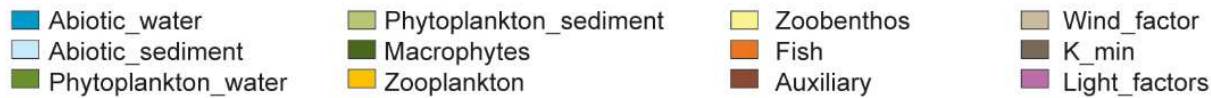


- **wind_factor (GOTM)**
- Factor to scale observed wind speed (in *meteo_file*)
- **A, g1, g2 (GOTM)**
- Light extinction coefficients
- Concentration of particles can also be important in simulating light conditions and therefore also temperature.

Andersen et al. (2020) *How morphology shapes the parameter sensitivity of lake ecosystem models*

Selecting important model parameters

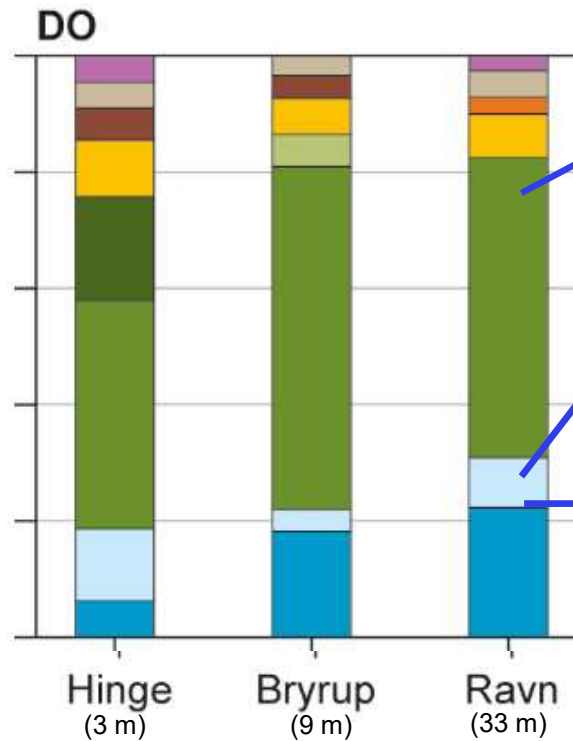
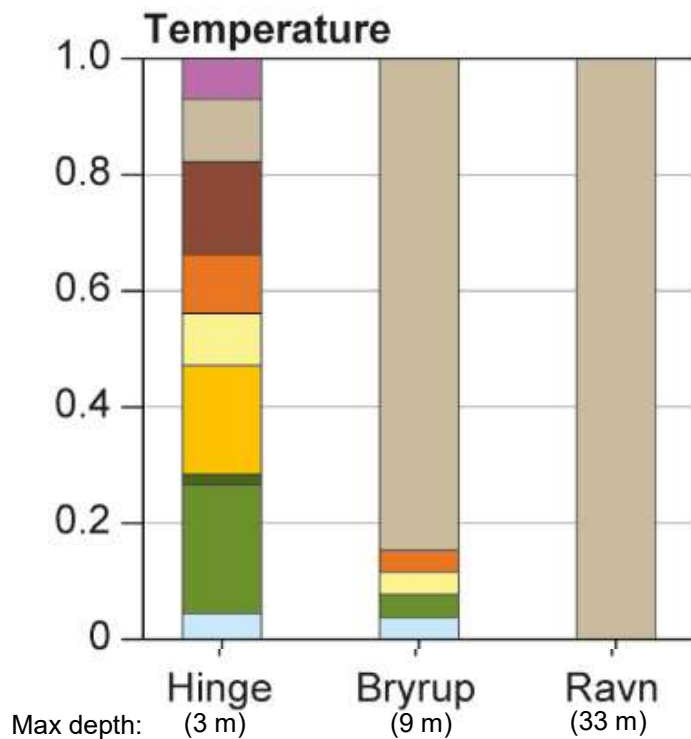
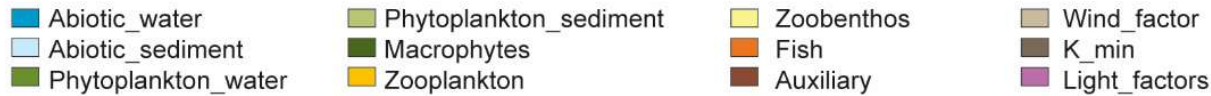
Andersen et al. (2020) *How morphology shapes the parameter sensitivity of lake ecosystem models*



- **wind_factor (GOTM)**
- Factor to scale observed wind speed (in *meteo_file*)
- **A, g1, g2 (GOTM)**
- Light extinction coefficients
- Concentration of particles can also be important in simulating light conditions and therefore also temperature.

Selecting important model parameters

Andersen et al. (2020) *How morphology shapes the parameter sensitivity of lake ecosystem models*



- **cMuMaxPhyto (WET)**
- Growth rate of phytoplankton groups
- **kDMinPOMS (WET)**
- Decomposition rate of POM-DW at 20°C
- **cThetaMinPOMS (WET)**
- Temperature coefficient of POM decomposition

Selecting important model parameters from SA

Subset of SA results from Andersen et al (2020) for Lake Ravn mean summer total chlorophyll a concentration:

Total chl-a											
cleaned						cleaned_sub1					
Parameter_name	S1	S1_conf	delta	delta_conf	module	Parameter_name	S1	S1_conf	delta	delta_conf	module
cFiltMax	0.199	0.008	0.168	0.0039	zooplankton	cFiltMax	0.2	0.0083	0.169	0.0045	zooplankton
cTmOptZoo	0.087	0.0059	0.115	0.0039	zooplankton	cTmOptZoo	0.087	0.0058	0.115	0.004	zooplankton
fDAssZoo	0.085	0.0056	0.097	0.0039	zooplankton	fDAssZoo	0.084	0.0059	0.097	0.0043	zooplankton
cPrefBlue	0.076	0.005	0.096	0.0039	zooplankton	cPrefBlue	0.076	0.0048	0.096	0.0044	zooplankton
cSigTmZoo	0.053	0.0046	0.086	0.0039	zooplankton	cSigTmZoo	0.053	0.0046	0.085	0.0041	zooplankton
cMuMaxBlue	0.044	0.0045	0.082	0.0041	phytoplankton_water	cMuMaxBlue	0.044	0.0041	0.083	0.004	phytoplankton_water



Selecting important model parameters from SA

Subset of SA results **UPDATED TO WET v2.1** for Lake Ravn mean summer total chlorophyll a concentration:

Total chl-a											
cleaned						cleaned_sub1					
Parameter_name	S1	S1_conf	delta	delta_conf	module	Parameter_name	S1	S1_conf	delta	delta_conf	module
kDConsMaxZoo	0.199	0.008	0.168	0.0039	zooplankton	kDConsMaxZoo	0.2	0.0083	0.169	0.0045	zooplankton
cTMOptZoo	0.087	0.0059	0.115	0.0039	zooplankton	cTMOptZoo	0.087	0.0058	0.115	0.004	zooplankton
fDAssZoo	0.085	0.0056	0.097	0.0039	zooplankton	fDAssZoo	0.084	0.0059	0.097	0.0043	zooplankton
-----	0.076	0.005	0.096	0.0039	zooplankton	-----	0.076	0.0048	0.096	0.0044	zooplankton
cSigTmZoo	0.053	0.0046	0.086	0.0039	zooplankton	cSigTmZoo	0.053	0.0046	0.085	0.0041	zooplankton
cMuMax	0.044	0.0045	0.082	0.0041	phytoplankton_water	cMuMax	0.044	0.0041	0.083	0.004	phytoplankton_water

NOTE: Some parameters have changed name from FABM-PCLake to WET v2.1.

If you cannot find parameter in fabm.yaml, it has likely been removed or renamed.

For instance, fish and zooplankton feeding parameters e.g. preference factors for phytoplankton (e.g. cPrefBlue) has been replaced with feeding parameters documented in Schnedler-Meyer & Andersen (2024, paper documenting fish in WET v2.1).

Understanding the ecosystem = Build parameter library

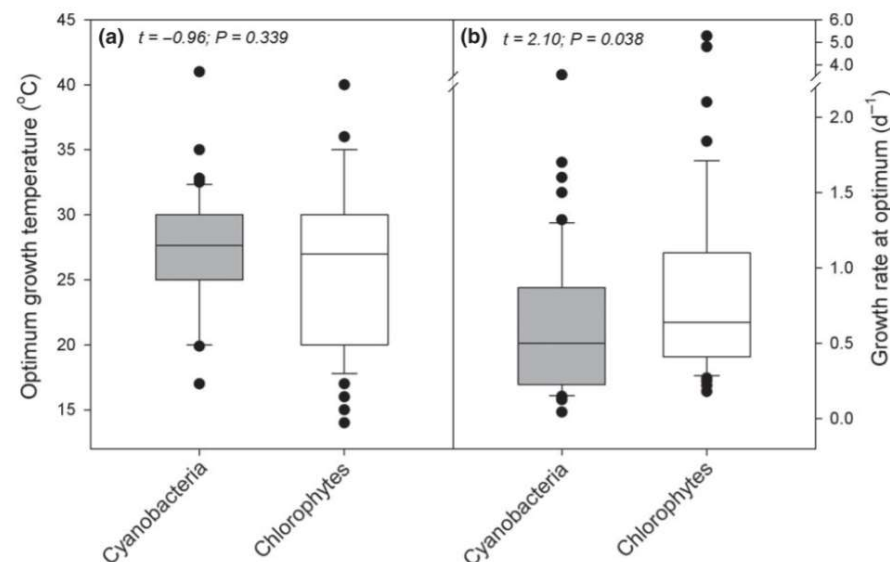
To understand your case study and to reproduce the ecosystem as best as possible with your model, you should **leverage as many relevant information sources as possible**, for instance scientific publications, reports or data sets/sources etc.

Journal of Freshwater Ecology 247

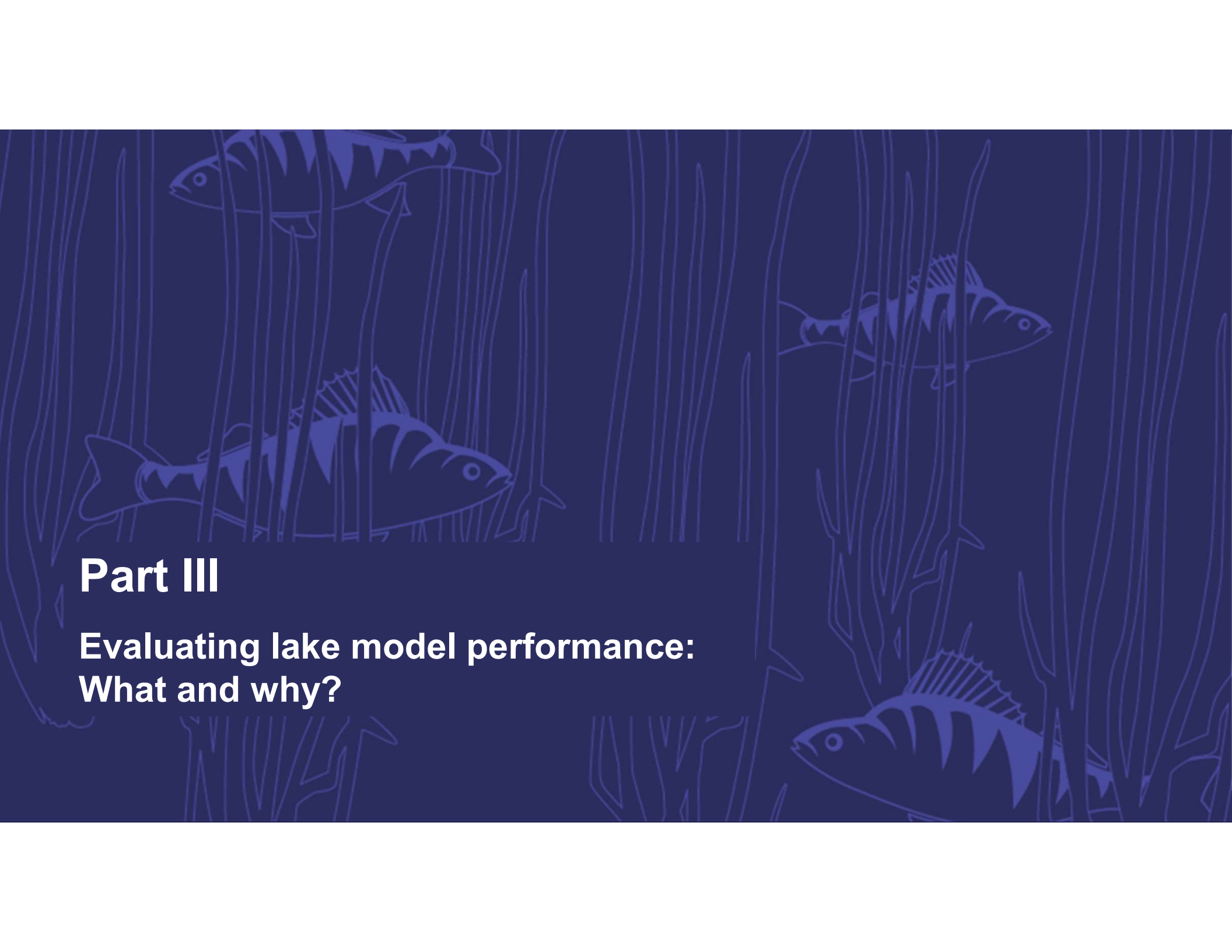
Table 3. The mean decomposition rates of the various organic phosphorus fractions for different decomposition models. The units of the decomposition rates are day⁻¹.

Rivers	Models and rate coefficients						<i>n</i>
	TOP → DIP	POP → DIP	POP → DOP → DIP		LPOP → LDOP → DIP		
	<i>k</i> _{TOP}	<i>k</i> _{POP1}	<i>k</i> _{POP2}	<i>k</i> _{DOP2}	<i>k</i> _{LPOP}	<i>k</i> _{LDOP}	
North Han	0.034	0.031	0.031	0.302	0.123	0.485	30
South Han	0.038	0.056	0.057	0.426	0.130	1.035	27
Lower Han	0.050	0.071	0.063	0.448	0.178	0.680	6
Kum	0.034	0.046	0.043	0.294	0.146	0.818	25
Youngsan	0.036	0.049	0.042	0.255	0.174	0.670	32
Sumjin	0.035	0.035	0.038	0.244	0.143	0.622	32
Mean (SD)	0.038 (0.006)	0.048 (0.013)	0.046 (0.011)	0.328 (0.080)	0.149 (0.021)	0.718 (0.172)	152

Islam et al. (2012). The decomposition rates of organic phosphorus and organic nitrogen in river waters. [Freshwater Biology](#)



Lürling et al. (2013). Comparison of cyanobacterial and green algal growth rates at different temperatures. [Freshwater Biology](#), 58(3), 552–559.

A stylized illustration of four striped bass swimming in a lake. The fish are depicted in a light blue color against a dark blue background. They are surrounded by vertical reeds or grasses, also in light blue. The fish are positioned at different depths and angles, giving a sense of movement. The overall style is minimalist and graphic.

Part III

**Evaluating lake model performance:
What and why?**

Model evaluation: The fundamentals

Model evaluation is always tightly linked to model purpose.

*“All models are approximations. Assumptions, whether implied or clearly stated, are never exactly true. **All models are wrong, but some models are useful.** So the questions you need to ask is not “Is the model true?” (it never is) but “Is the model good enough for this particular application?””*

George Box

Modelers need to ask:

“What makes the model fit-for-purpose?”

Model evaluation: The fundamentals

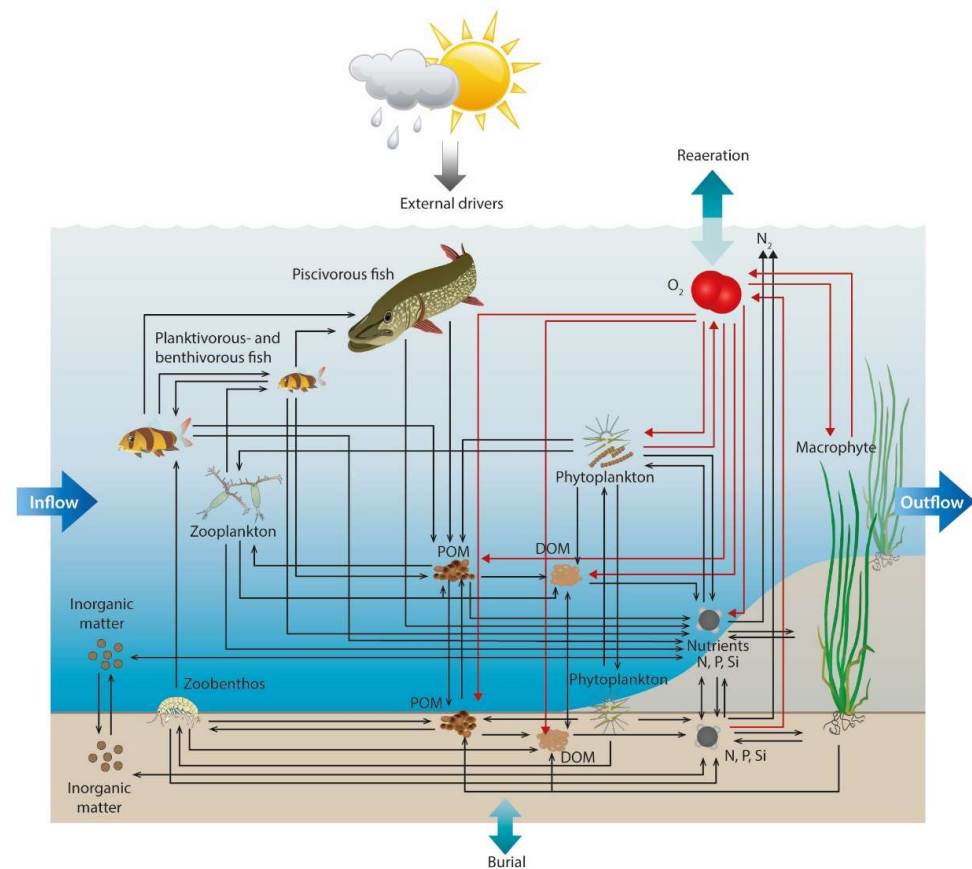
Model evaluation is always tightly linked to model purpose.

*“All models are approximations. Assumptions, whether implied or clearly stated, are never exactly true. **All models are wrong, but some models are useful.** So the questions you need to ask is not “Is the model true?” (it never is) but “Is the model good enough for this particular application?””*

George Box

Modelers need to ask:

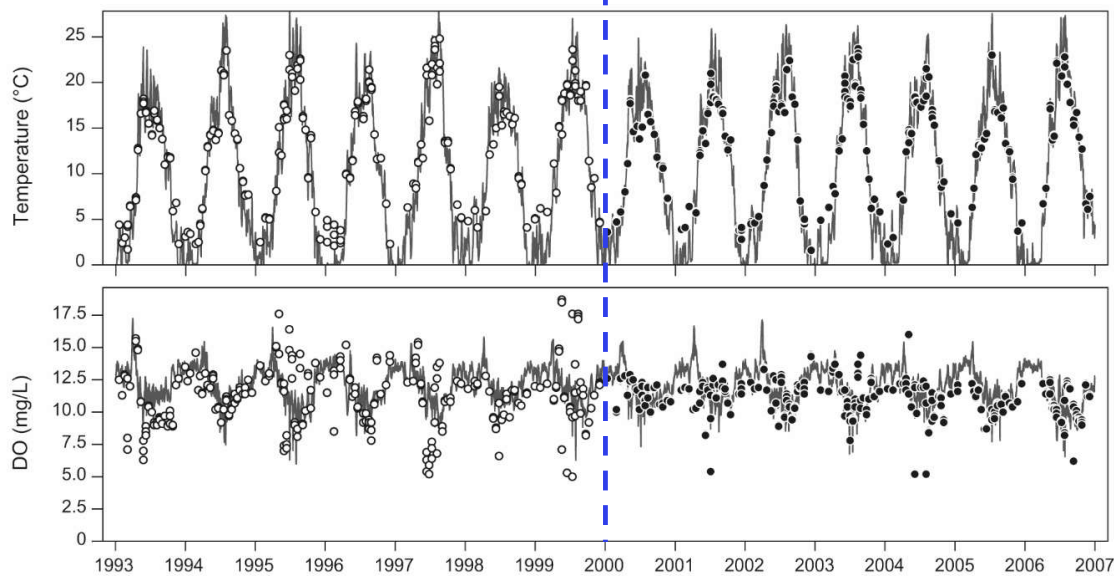
“What makes the model fit-for-purpose?”



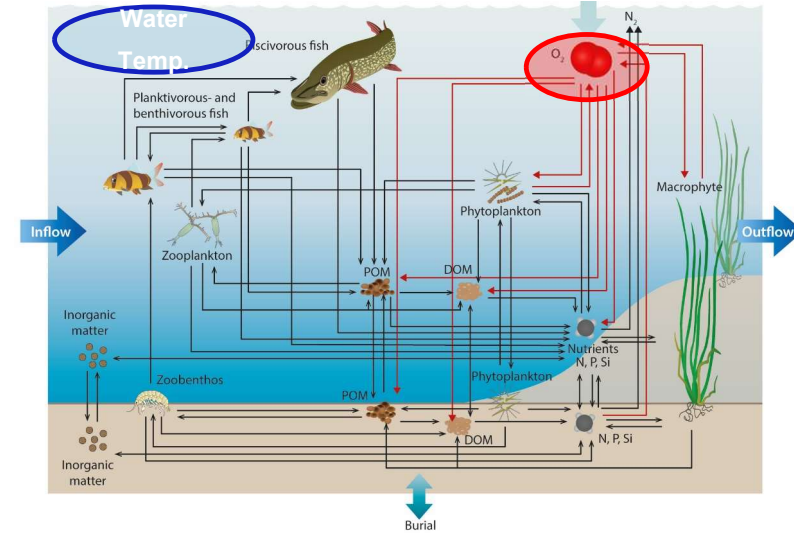
WQ state variables

Calibration: 1993-2000

Validation: 2000-2007



Andersen et al, 2020, *Ecol. Appl.*



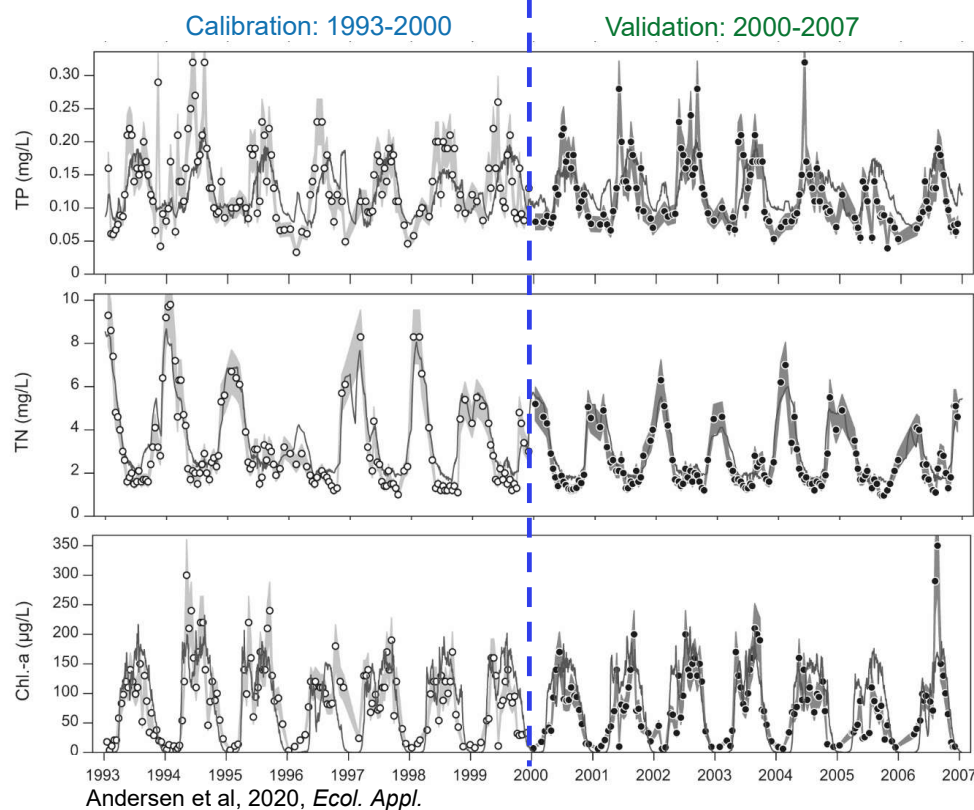
R^2 : 0.96|0.98

BIAS (%): -2.0| 2.3

R^2 : 0.18|0.21

BIAS (%): 6.0| 5.1

WQ state variables



R^2 : 0.24|0.32

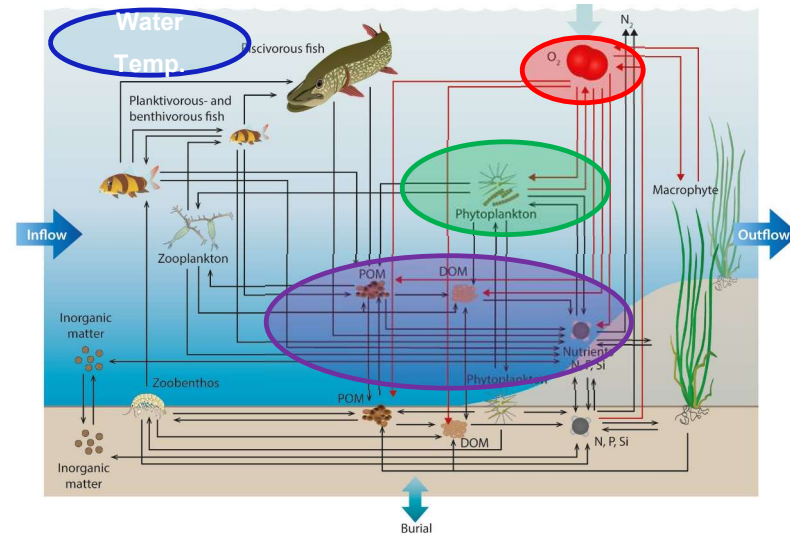
BIAS (%): -5.1| 8.0

R^2 : 0.89|0.83

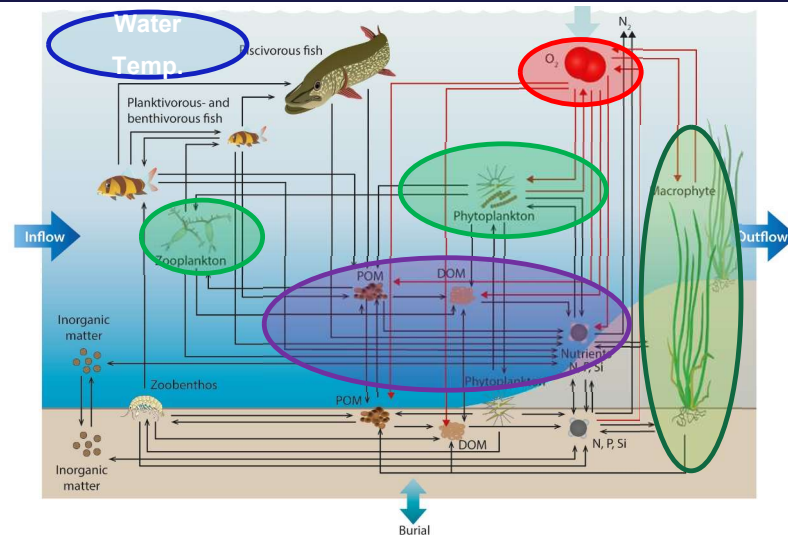
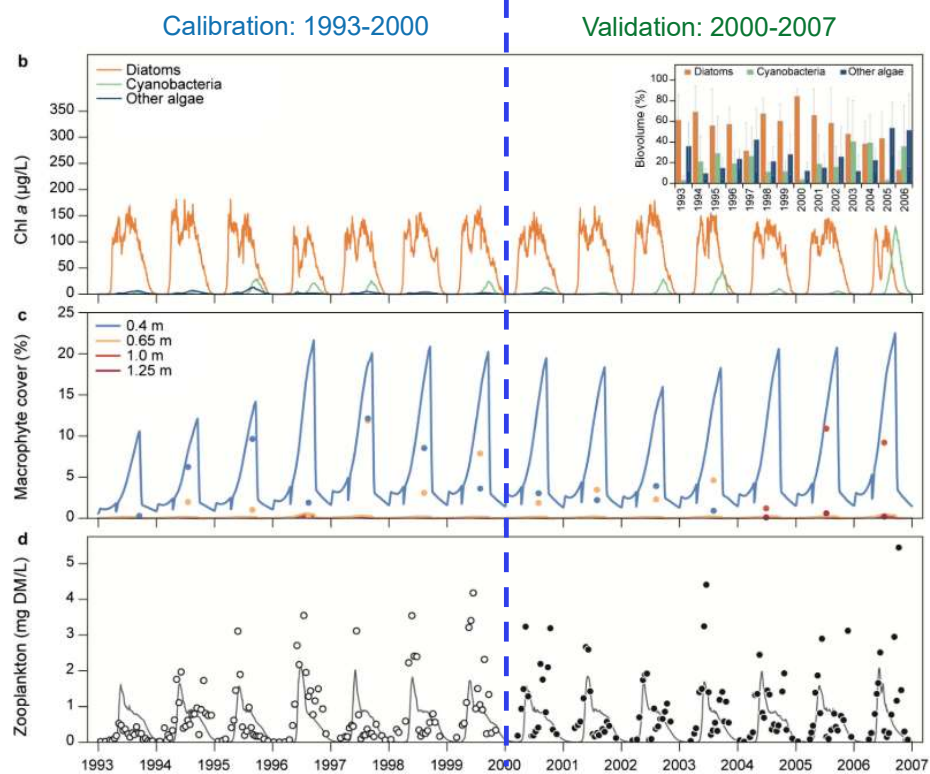
BIAS (%): 10.8| 9.7

R^2 : 0.43|0.43

BIAS (%): -6.6| 6.0

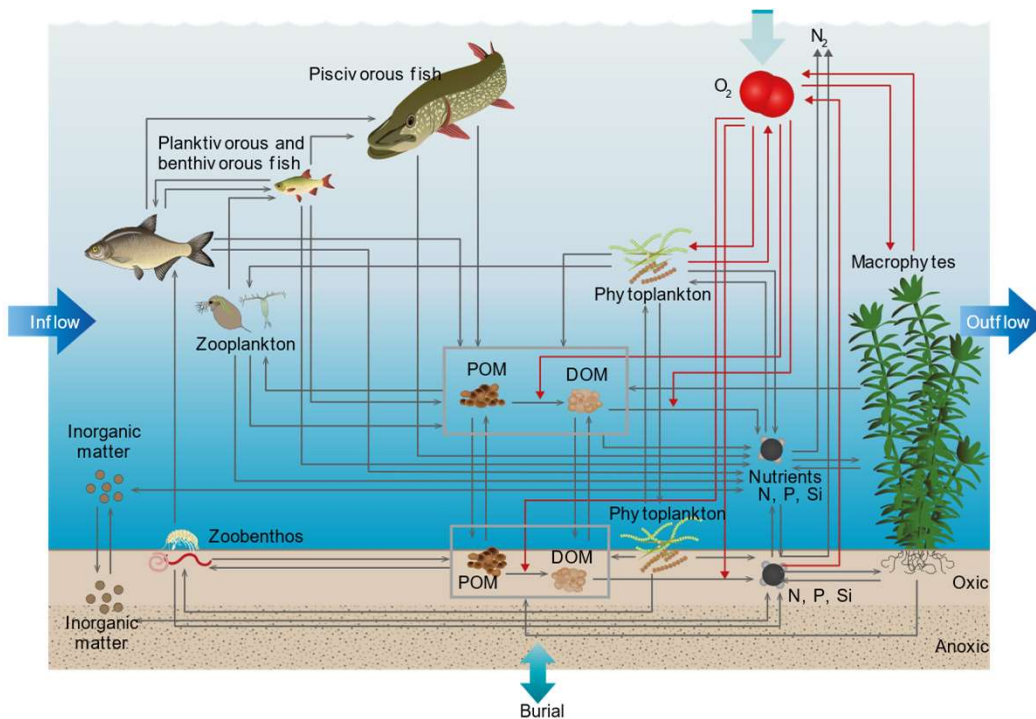


WQ state variables

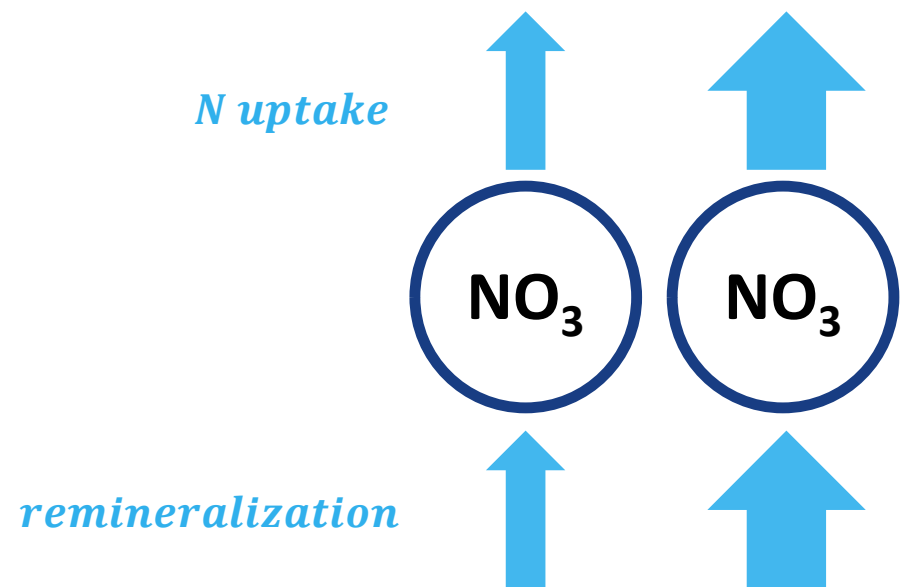


Model evaluation: The fundamentals

How do we best evaluate lake ecosystem models?



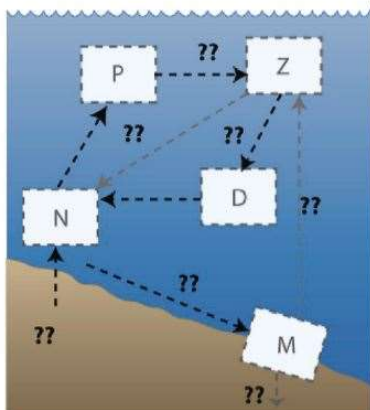
$$\frac{dN}{dt} = \text{remineralization} - N \text{ uptake}$$



Beyond state variables: CSPC framework

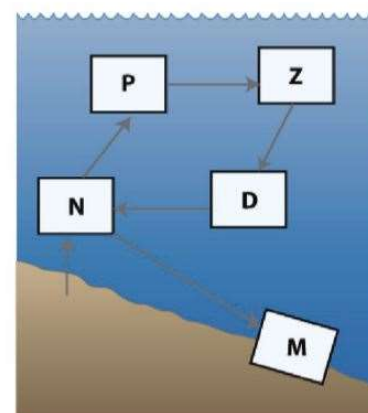
LEVEL 0: CONCEPTUAL VALIDATION

Validation focus:
Is the model configured appropriately?
Is their sound empirical basis for assumptions
about model structure and parameterisation?



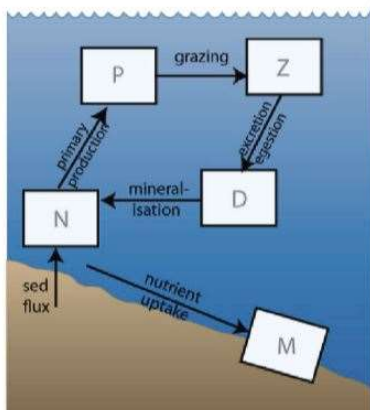
LEVEL 1: STATE VALIDATION

Validation focus:
Do simulated variables match observed levels?
Does the model capture variability over
a relevant range of spatio-temporal scales?



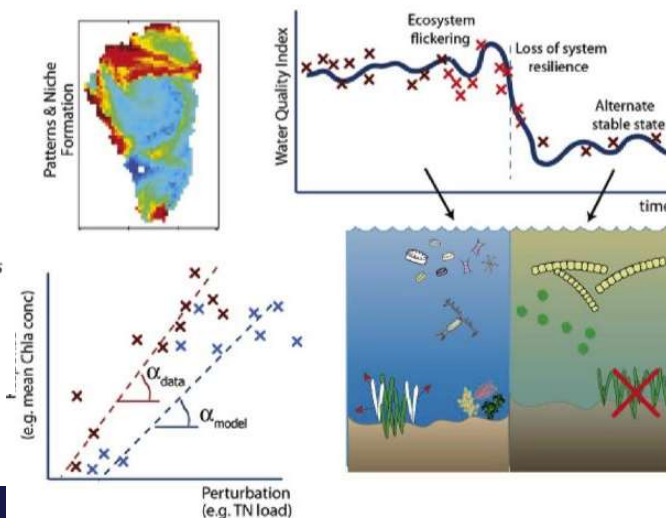
LEVEL 2: PROCESS VALIDATION

Validation focus:
Process validation to reduce equifinality
Are simulated rates and fluxes accurate?



LEVEL 3: SYSTEM VALIDATION

Validation focus:
Emergent patterns & signatures
System-scale perturbation-response relationships
Resilience and ecosystem stability
Ecosystem services

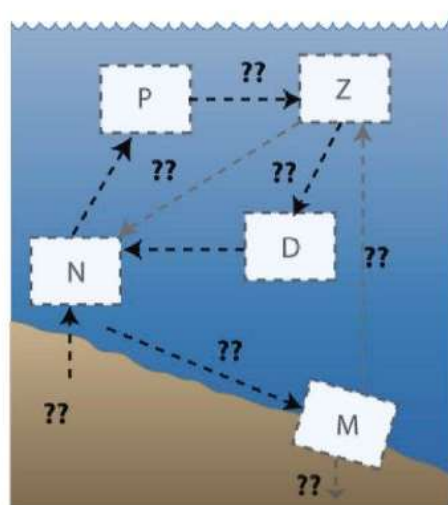


Beyond state variables: CSPC framework

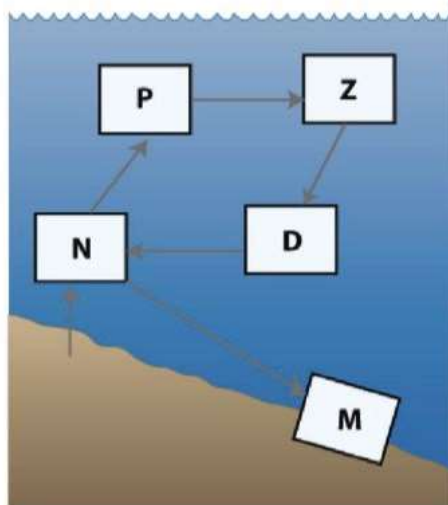
Structure → Process → Function

Improved predictability

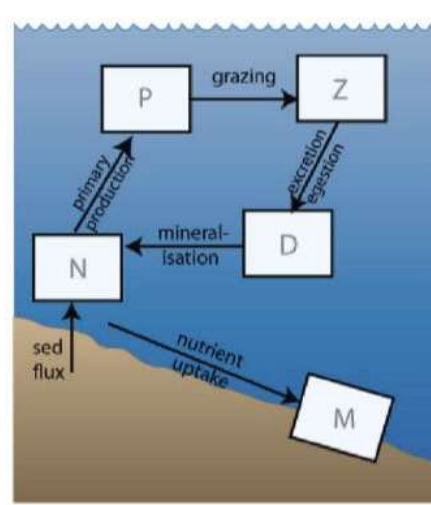
Reduced uncertainty



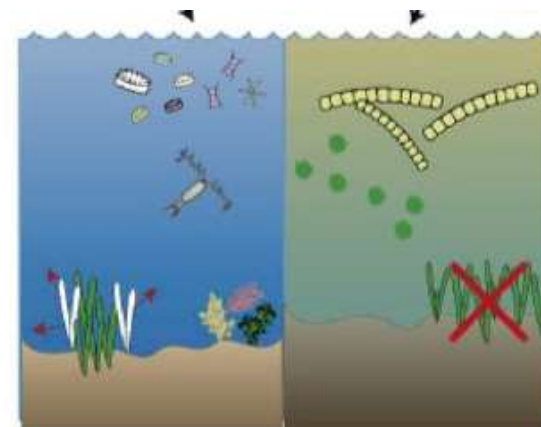
Level 0:
Conceptual validation



Level 1:
State validation



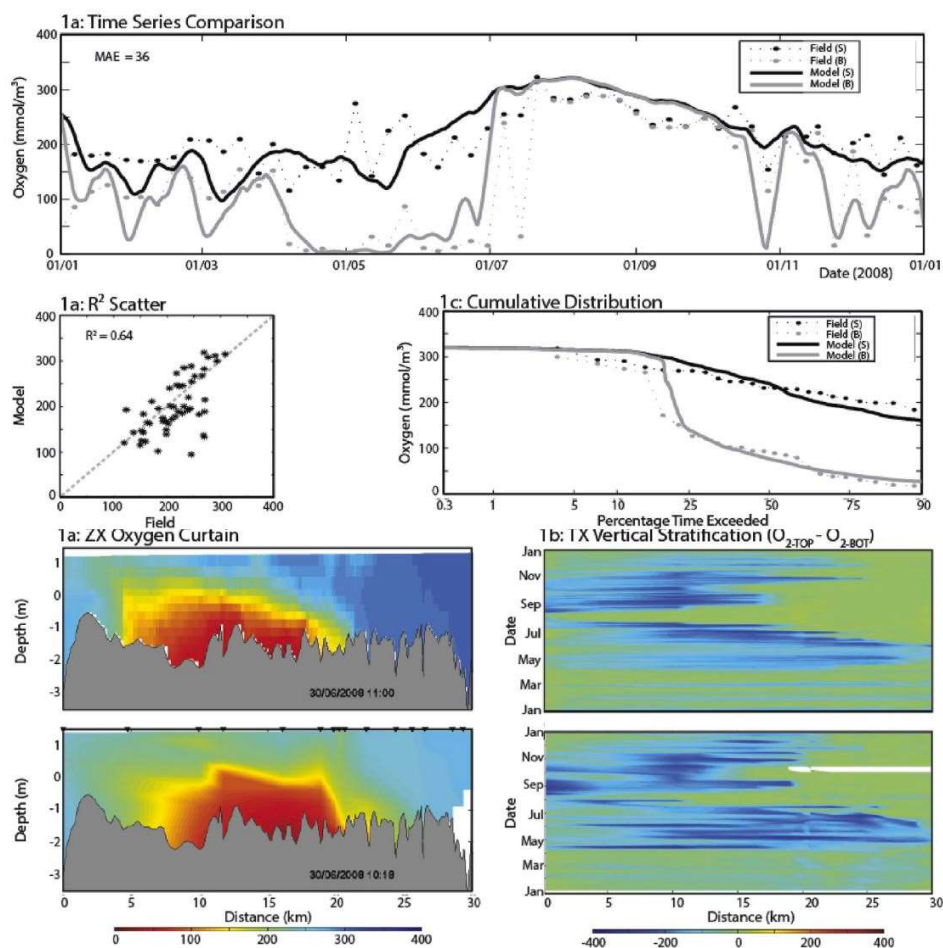
Level 2:
Process validation



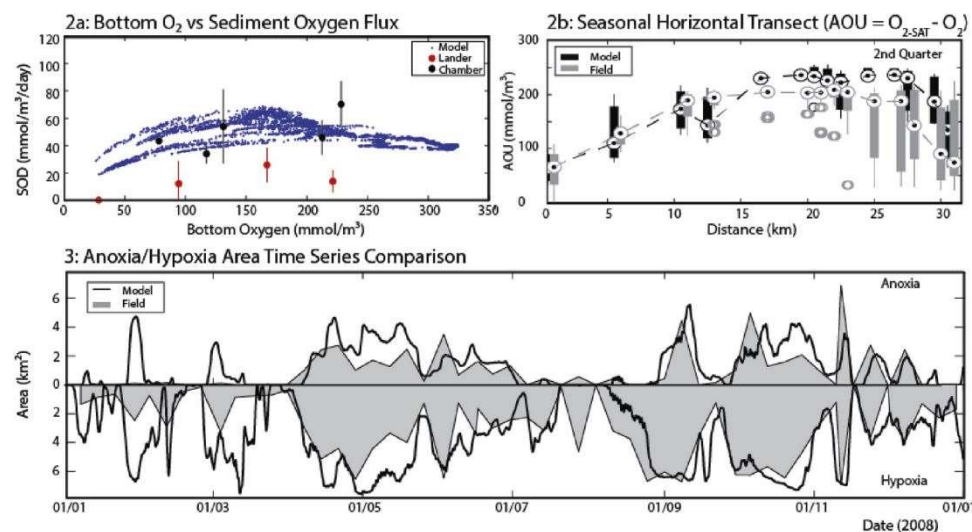
Level 2:
System validation

CSPS framework: an example

Level 1: State



Level 2: Process



Level 3: System

CSPS framework: a WET example

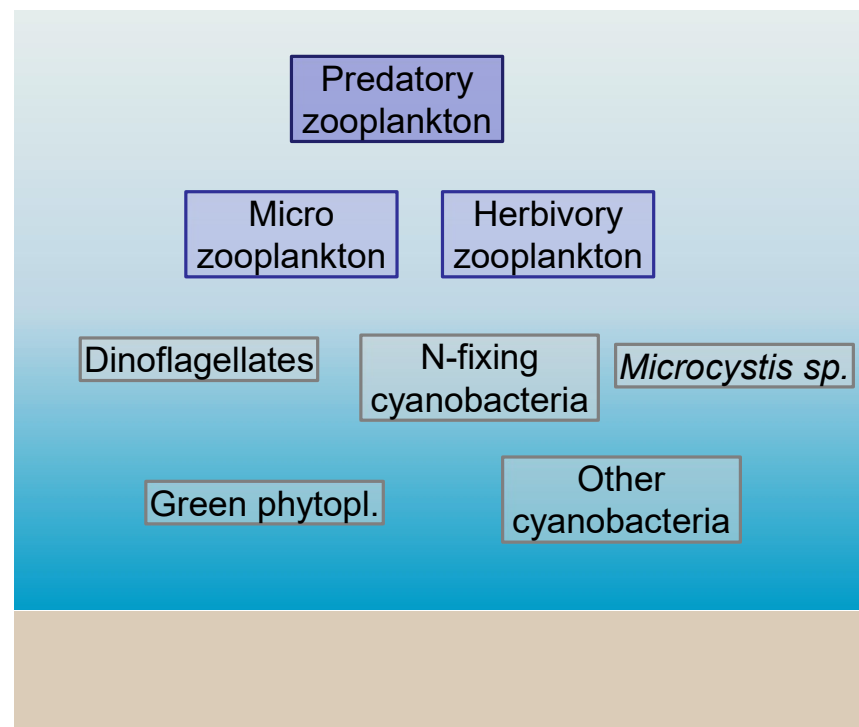
Regev, Carmel, Gal (2023)
Environ. Model. Softw.

Modelling Lake Kinneret (Sea of Galilee), a warm monomictic lake

Level 0: Conceptual

Table 3
Conceptual validation questions set forth by [Hipsey et al. \(2020\)](#) and responses.

Questions	Responses
(1) Conceptual representation	
(1 b) Does the model structure reflect recent advances in ecosystem understanding that may be important in this application?	Some ecosystem elements, such as anammox and dissimilative reduction of nitrate to ammonia (DRNA) processes (Madigan et al., 2012), were not incorporated in the WET model. Adding processes that we have little information about, will increase uncertainty. Also, simplicity was favored for accuracy.
(1c) Does the model reflect the system understanding of relevant local disciplinary experts and stakeholders?	Yes, the model setup was based on the DYRESM-CAEDYM setup (Gal et al., 2009) with some modification, and was build according to the decision makers questions and needs.



CSPS framework: a WET example

Regev, Carmel, Gal (2023)
Environ. Model. Softw.

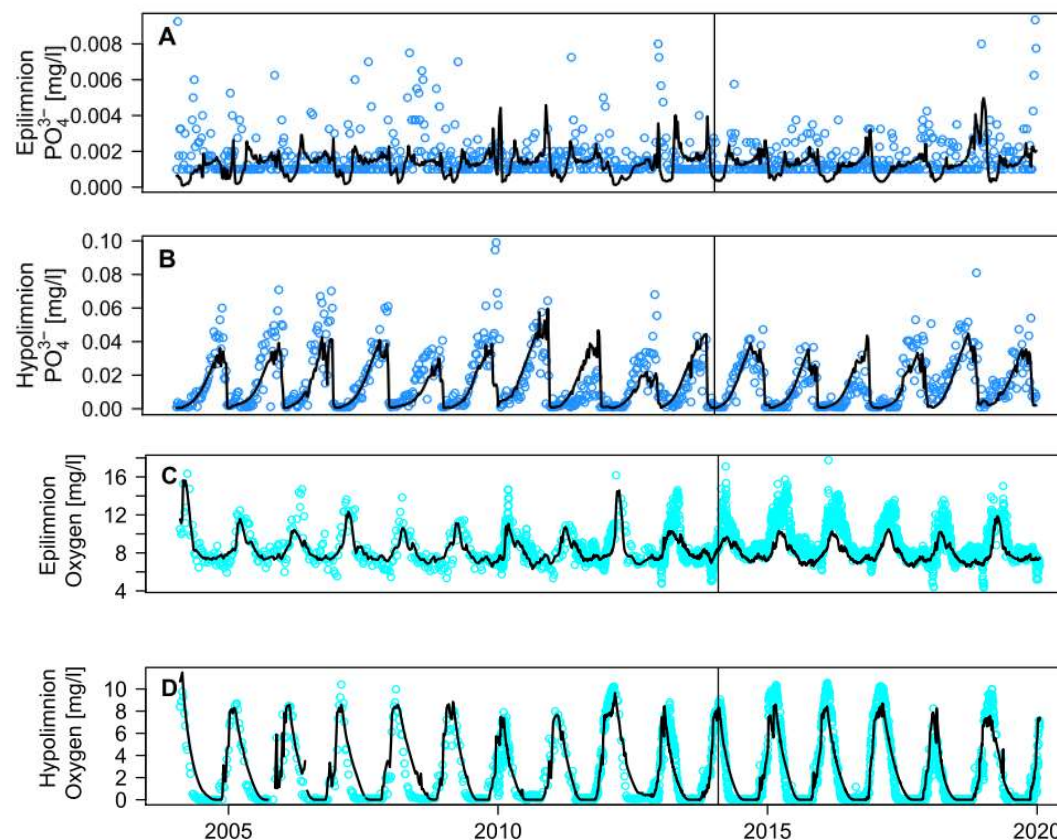
Modelling Lake Kinneret (Sea of Galilee), a warm monomictic lake

Level 0: Conceptual

Table 3
Conceptual validation questions set forth by [Hipsey et al. \(2020\)](#) and responses.

Questions	Responses
(1) Conceptual representation	
(1 b) Does the model structure reflect recent advances in ecosystem understanding that may be important in this application?	Some ecosystem elements, such as anammox and dissimilative reduction of nitrate to ammonia (DRNA) processes (Madigan et al., 2012), were not incorporated in the WET model.
	Adding processes that we have little information about, will increase uncertainty. Also, simplicity was favored for accuracy.
(1c) Does the model reflect the system understanding of relevant local disciplinary experts and stakeholders?	Yes, the model setup was based on the DYRESM-CAEDYM setup (Gal et al., 2009) with some modification, and was build according to the decision makers questions and needs.

Level 1: State

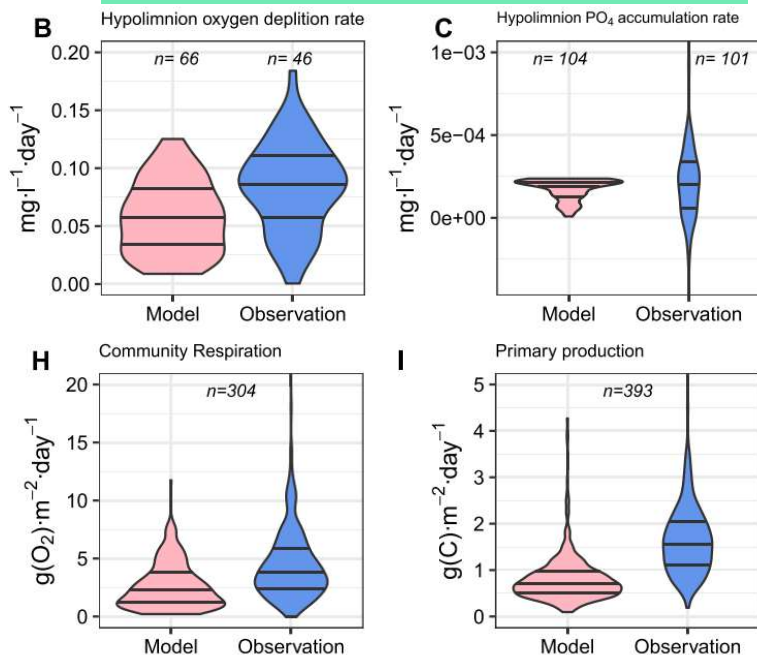


CSPS framework: a WET example

Regev, Carmel, Gal (2023)
Environ. Model. Softw.

Modelling Lake Kinneret (Sea of Galilee), a warm monomictic lake

Level 2: Process



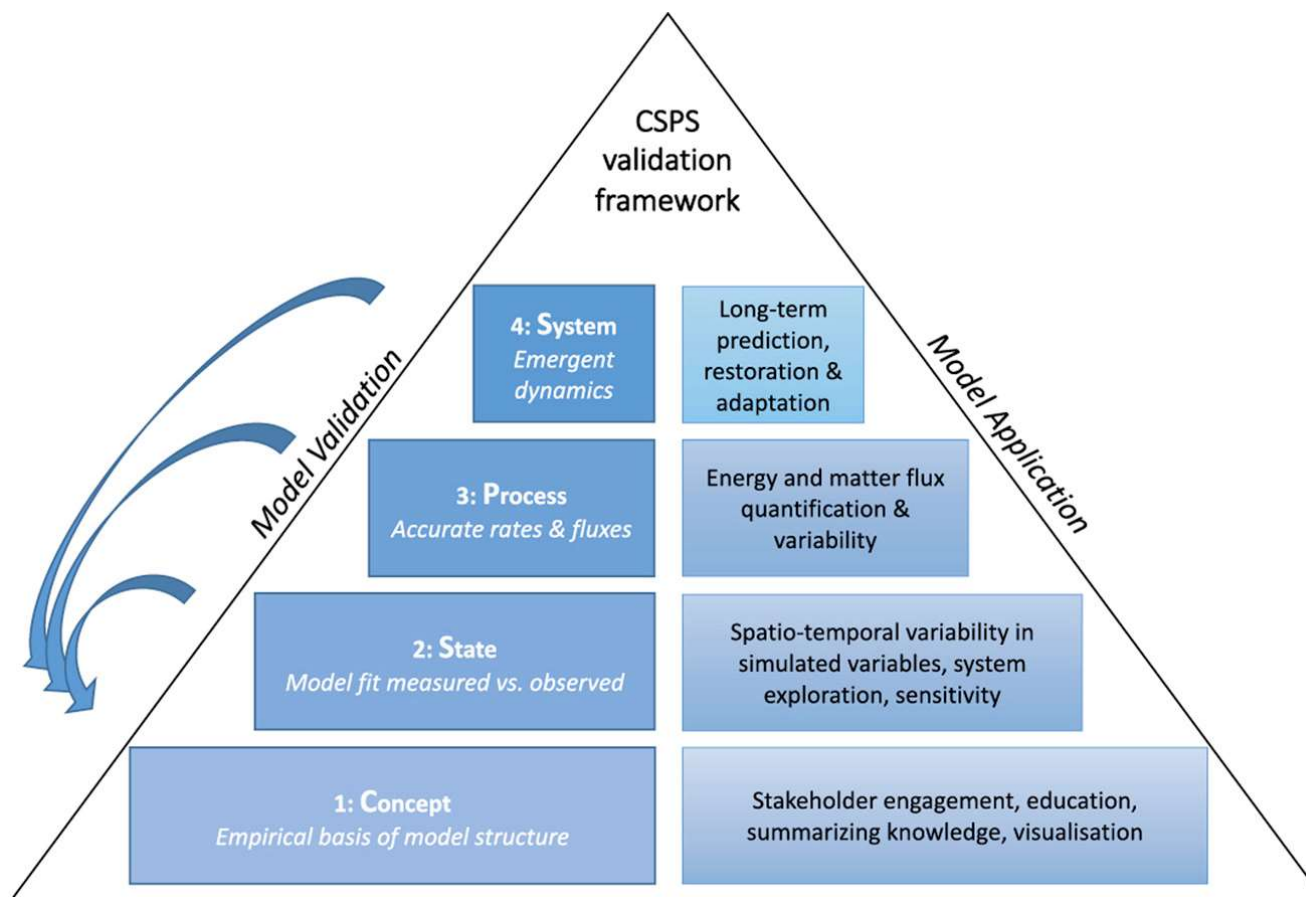
Level 3: System

Table 6

System Validation. R^2 , P-val and slope of relation between variables in observation versus model. Based on epilimnion average monthly data, 30 years of observation and 16 years of model run. Abbreviation used: Tilda sign (~) denote linear relation tested between 2 variables; lv_diff = lake level difference between consecutive months; $NDOM/NPOM_Jordan$ = monthly mean of N dissolved/particulate organic matter concentration as measured in the Jordan stream multiplied by the flow expressed in [g/s]. In case a non-linear relation, the data were log transformed. Relations are interpreted as “agreement” if the linear model slope had same sign for both observation and model, and “disagreement” otherwise.

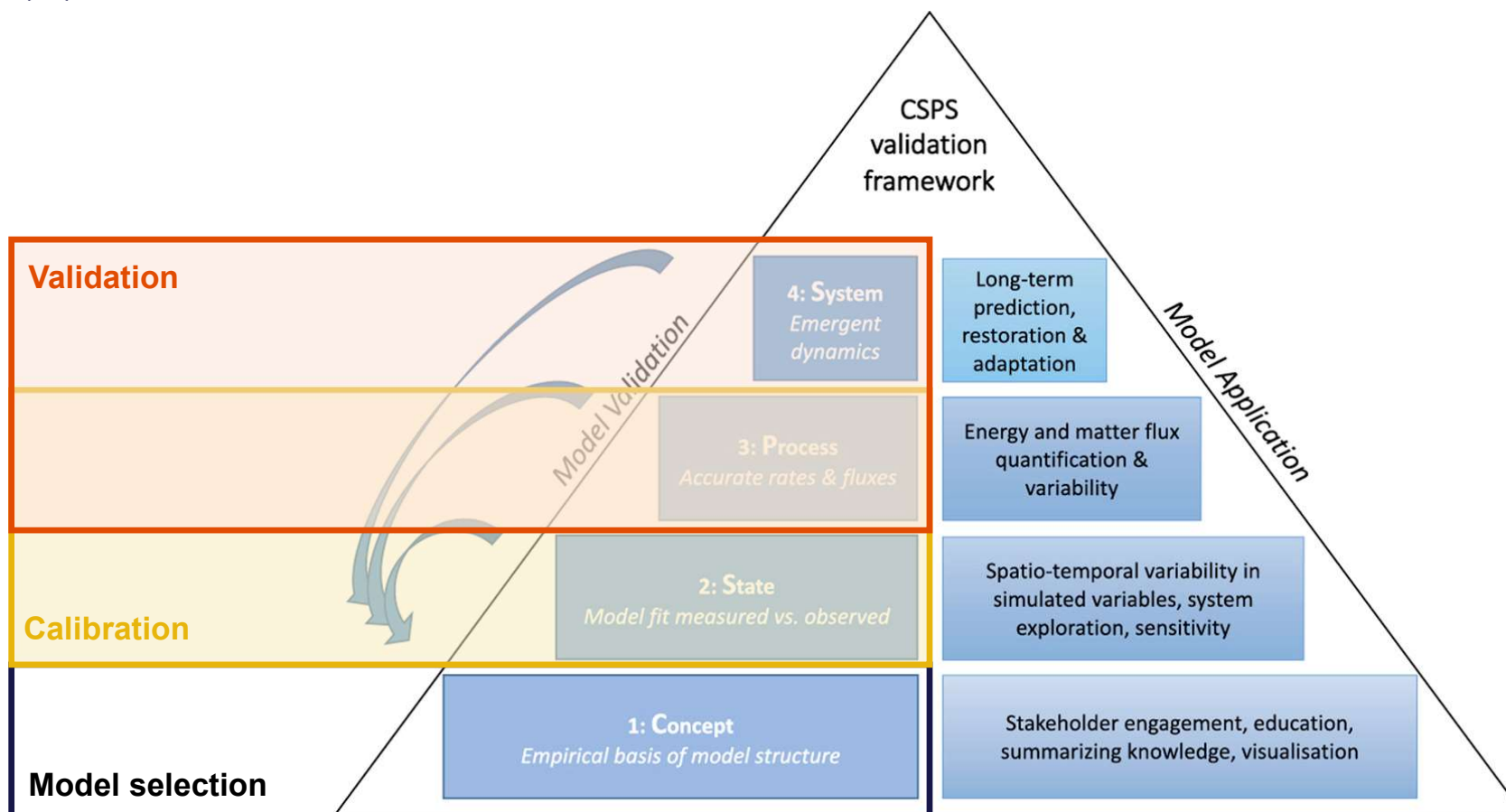
Relation	n		R^2		Slope		Results interpretation
	Obs	Model	Obs	Model	Obs	Model	
$lv_diff \sim NO_3$	356	194	0.6	0.46	0.192	0.258	Agreement
$Dino \sim NPOM$	356	194	0.52	0.06	64	36	Agreement
$lv_diff \sim Temp$	356	194	0.51	0.52	-0.048	-0.046	Agreement
$lv_diff1 \sim O_2$	356	194	0.47	0.58	0.126	0.184	Agreement
$lv_diff1 \sim \log(NfixCyano)$	255	194	0.41	0.36	-0.082	-0.182	Agreement
$MicroZoo \sim HerbZoo$	356	194	0.39	0.21	0.293	-0.77	Disagreement
$lv_diff1 \sim \log(Dino)$	356	194	0.37	0.41	0.18	0.065	Agreement
$\log(Dino) \sim O_2$	356	194	0.36	0.37	0.38	1.46	Agreement
$\log(NfixCyano) \sim \log(NO_3)$	255	194	0.35	0.39	-0.86	-1.11	Agreement
$NO_3 \sim O_2$	356	194	0.35	0.16	0.44	0.268	Agreement
$NO_3 \sim NDOM_Jordan_mean$	295	179	0.34	0.45	0.017	0.018	Agreement
$\log(NfixCyano) \sim Temp$	255	194	0.32	0.58	0.283	0.16	Agreement
$\log(Dino) \sim \log(NPOM_Jordan_mean)$	340	181	0.29	0.43	0.66	2.2	Agreement
$\log(Dino) \sim \log(NfixCyano)$	255	194	0.28	0.05	-0.217	-0.69	Agreement
$MicroZoo \sim PredZoo$	356	194	0.26	0.74	1.58	1.05	Agreement

CSPS framework and model application



Hipsey et al (2020) *Environ. Mod. Softw.*

CSPS framework and calibration/validation

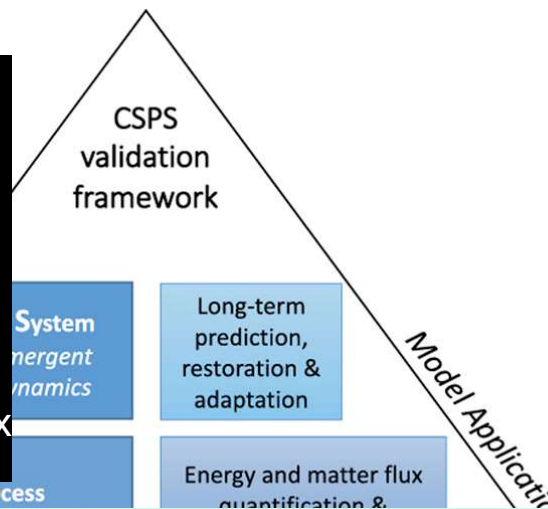


Hipsey et al (2020) *Environ. Mod. Softw.*

CSPS framework and model application

*"All models are approximations. Assumptions, whether implied or clearly stated, are never exactly true. **All models are wrong, but some models are useful.** So the questions you need to ask is not "Is the model true?" (it never is) but "Is the model good enough for this particular application?"*

George Box



You need to understand or hypothesize about the fundamental ecological processes in your lake ecosystem

Take aways

DTU Calibration and validation

8. Calibration and validation

"Calibration is adjusting model parameters with the purpose of achieving the best simulation match with observations."

This may be achieved by a combination of visual inspection and optimization of an objective function.

For example, minimizing the Mean Absolute Percentage Error (MAPE):

$$MAPE(\text{Chl}) = \frac{1}{n} \sum_{i=1}^n \frac{|Chl_{model,i} - Chl_{obs,i}|}{Chl_{obs,i}}$$

8. Calibration and validation

"Validation is the process of testing the calibrated parameters with an independent set of data (in time and/or space)"

Model adjustment to Lake Hinge case

Andersen et al. (2010) *Sci. Rep.*

DTU Beyond state variables: CSPC framework

Structure → Process → Function

Improved predictability
Reduced uncertainty

Level 0: Conceptual validation
Level 1: State validation
Level 2: Process validation
Level 3: System validation

Hansen et al. (2010) *Environ. Mod. Soft.*

DTU Calibration strategy for WET

Step	Model focus	Processes	Observations
1	GOTM	Water column mixing	Water temperature
2	Metabolism	Decomposition and mineralization	Dissolved oxygen
3	Nitrogen	Denitrification and nitrification	NO ₃ and NH ₄
4	Phosphorous	P sediment release	PO ₄
5	Plankton	Nutrient uptake and grazing	TP, TN, Chl-a
6	Others		

DTU Selecting important model parameters

Andersen et al. (2011) *How morphology shapes the parameter sensitivity of lake ecosystem models*

- cMuMaxPhyto (WET)
- Growth rate of phytoplankton groups
- kDMinPOMs (WET)
- Decomposition rate of POM-DW at 20°C
- cThetaMinPOMs (WET)
- Temperature coefficient of POM decomposition